

E.G.S. PILLAY ENGINEERING COLLEGE

(Autonomous)

NAGAPATTINAM – 611002

(Affiliated to Anna University, Chennai | Accredited by NAAC with „A++“
Grade | Accredited by NBA T1 (B.E. – CSE, CIVIL, ECE, EEE, MECH & B.Tech – IT) |
Approved by AICTE, New Delhi)



MASTER OF COMPUTER APPLICATIONS R - 2024

SECOND YEAR

CURRICULUM AND SYLLABUS FOR THIRD SEMESTER

SEMESTER III									
CourseCode	CourseName	L	T	P	C	MaximumMarks			Category
						CIA	ES	Total	
Theory Course									
2402CA301	Machine Learning	3	-	-	3	40	60	100	PC
2402CA302	Internet of Things	3	-	-	3	40	60	100	PC
	Open elective	3	-	-	3	40	60	100	OE
	Elective – II	3	-	-	3	40	60	100	PE
	Elective – III	3	-	-	3	40	60	100	PE
	Elective – IV	3	-	2	4	50	50	100	PE
Laboratory Course									
2402CA303	Machine Learning Laboratory	-	-	4	2	60	40	100	PC
2402CA304	Internet of Things Laboratory	-	-	4	2	60	40	100	PC
2404CA305	Professional Development Course III	-	-	2	1	100	-	100	EEC
Total		18	0	12	24	470	430	900	

2402CA301

MACHINE LEARNING

L T P C
3 0 0 3

Prerequisite Data Mining and Data Warehousing

COURSE OBJECTIVES

1. To gain knowledge on foundations of machine learning and apply suitable dimensionality reduction techniques for an application
2. To select the appropriate model and use feature engineering techniques
3. To gain knowledge on Probability and Bayesian Learning to solve the given problem
4. To design and implement the machine learning techniques for real world problems

COURSE OUTCOMES

At the end of the course, students will be able to

- CO1 Apply data preprocessing and dimensionality reduction techniques to real datasets for model preparation
- CO2 Apply appropriate machine learning models and feature engineering techniques for a given problem
- CO3 Utilize probabilistic techniques and Bayesian learning methods to solve given problems.
- CO4 Apply and analyze parametric models like logistic regression and neural networks to solve classification problems and address overfitting
- CO5 Choose and apply appropriate machine learning algorithms to learn and classify data effectively

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	2	–	3	–	–	–	–	2	3
CO2	3	3	3	3	–	–	–	–	3	3
CO3	3	3	3	2	–	–	–	–	3	3
CO4	3	3	3	3	–	–	–	–	3	3
CO5	3	3	3	3	2	–	–	–	3	3
Avg	3	2.8	3	2.8	2	–	–	–	2.8	3

UNIT I INTRODUCTION

9 Hours

Human Learning - Types – Machine Learning - Types - Problems not to be solved - Applications - Languages/Tools– Issues. Preparing to Model: Introduction - Machine Learning Activities - Types of data - Exploring structure of data - Data quality and remediation - Data Pre-processing

UNIT II UNSUPERVISED LEARNING AND REINFORCEMENT LEARNING, FEATURE ENGINEERING

9 Hours

Introduction - Clustering Algorithms -K – Means – Hierarchical Clustering - Cluster Validity - Dimensionality Reduction –Linear Discriminant Analysis-Principal Component Analysis- Reinforcement Learning : Q learning, Rewards and Actions Temporal Difference Learning - Markov Decision Process. Feature Engineering: Feature Transformation - Feature Subset Selection

UNIT III BAYESIAN LEARNING

8 Hours

Basic Probability Notation- Inference – Independence - Bayes' Rule. Bayesian Learning: Maximum Likelihood and Least Squared error hypothesis-Maximum Likelihood hypotheses for predicting probabilities- Minimum description Length principle -Bayes optimal classifier - Naïve Bayes classifier - Bayesian Belief networks -EM algorithm- Jupyter Notebook

UNIT IV PARAMETRIC MACHINE LEARNING

9 Hours

Logistic Regression: Classification and representation – Cost function – Gradient descent – Advanced optimization – Regularization - Solving the problems on overfitting. Perceptron – Neural Networks – Multi –

class Classification - Backpropagation – Non-linearity with activation functions (Tanh, Sigmoid, Relu, PRelu) - Dropout as regularization

UNIT V NON PARAMETRIC MACHINE LEARNING

10 Hours

k- Nearest Neighbors- Decision Trees – Branching – Greedy Algorithm - Multiple Branches – Continuous attributes – Pruning. Random Forests: ensemble learning. Boosting – Adaboost algorithm. Support Vector Machines – Large Margin Intuition – Loss Function - Hinge Loss – SVM Kernels

TOTAL : 45 PERIODS

1.Ethem Alpaydin, “Introduction to Machine Learning 3e (Adaptive Computation and Machine Learning Series)”, Third Edition, MIT Press, 2014

2.Tom M. Mitchell, “Machine Learning”, India Edition, 1st Edition, McGraw-Hill Education Private Limited, 2013

3.Saikat Dutt, Subramanian Chandramouli and Amit Kumar Das, "Machine Learning", 1st Edition, Pearson Education, 2019

4.Christopher M. Bishop, “Pattern Recognition and Machine Learning”, Revised Edition, Springer, 2016.

5.Aurelien Geron, “Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow”, 2nd Edition, O’Reilly, 2019

2402CA302

INTERNET OF THINGS

L T P C
3 0 0 3

Prerequisite Computer Communication and Network ,Python Programming

COURSE OBJECTIVES

1. To understand the concepts of IoT and its working models
2. To know the various IoT protocols
3. To understand about various IoT Physical devices and Endpoints
4. To know the security and privacy issues connected with IoT
5. To apply the concept of Internet of Things in a real world scenario.

COURSE OUTCOMES

- On the successful completion of the course, students will be able to
- CO1 Apply the fundamental infrastructure and architecture required for IoT deployments.
- CO2 Apply IoT communication protocols for interaction between devices and networks.
- CO3 Design and implement IoT applications using Arduino, Raspberry Pi, or equivalent platforms.
- CO4 Explain the concepts of security, privacy, and governance in IoT systems.
- CO5 Analyze and evaluate IoT applications in real-world scenarios.

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	2	2	2	1	-	-	1	3	2
CO2	2	3	2	3	1	-	-	1	3	3
CO3	2	3	3	3	2	2	-	1	3	2
CO4	1	2	2	2	-	-	3	1	2	1
CO5	2	3	3	3	2	1	1	3	3	3
Avg	2	2.6	2.4	2.6	1.5	1.5	2	1.4	2.8	2.2

UNIT I FUNDAMENTALS OF IOT

9 Hours

Definition and Characteristics of IoT, Sensors, Actuators, Physical Design of IoT – IoT Protocols, IoT communication models, IoT Communication APIs, IoT enabled Technologies –Wireless Sensor Networks, CloudComputing, Embedded Systems, IoT Levels and Templates, Domain Specific IoTs –Home, City, Environment, Energy, Agriculture and Industry

UNIT II IOT PROTOCOLS

9 Hours

Protocol Standardization for IoT – Efforts – M2M and WSN Protocols – SCADA and RFID Protocols – Issues with IoT Standardization – Unified Data Standards – Protocols – IEEE802.15.4–BACNet Protocol– Modbus – KNX – Zigbee– Network layer – APS layer – Security

UNIT III IOT PHYSICAL DEVICES AND END POINTS

9 Hours

Introduction to Arduino and Raspberry Pi- Installation, Interfaces (serial, SPI, I2C), Programming – Python program with Raspberry PI with focus on interfacing external gadgets, controlling output, and reading input from pins.

UNIT IV INTERNET OF THINGS PRIVACY, SECURITY AND GOVERNANCE

9 Hours

Introduction, Overview of Governance, Privacy and Security Issues, Contribution from FP7 Projects, Security, Privacy and Trust in IoT-Data-Platforms for Smart Cities, First Steps Towards a Secure Platform, Smartie Approach. Data Aggregation for the IoT in Smart Cities, Security

UNIT V CASE STUDIES AND ADVANCED TOPICS

9 Hours

IOT APPLICATIONS-IoT applications for industry: Future Factory Concepts, Brownfield IoT, Smart Objects, Smart Applications. Study of existing IoT platforms/middleware, IoT-A, Hydra etc.

REFERENCE

TOTAL : 45 PERIODS

Internet of Things - A Hands-on Approach, Arshdeep Bahga and Vijay Madisetti, Universities
Olivier Hersent, David Boswarthick, Omar Elloumi, "The Internet of Things – Key applications and Protocols", Wiley, 2012.
David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Rob Barton, Jerome Henry, "IoT Fundamentals, Networking Technologies, Protocols, and Use cases for the Internet of Things", Cisco Press, First Edition, 2017.
Dieter Uckelmann, Mark Harrison, Michahelles, Florian (Eds), "Architecting the Internet of Things", Springer, 2011
Raspberry Pi Cookbook, Software and Hardware Problems and solutions, Simon Monk, O'Reilly (SPD), 2016, ISBN 7989352133895
Peter Friess, 'Internet of Things – From Research and Innovation to Market Deployment', River Publishers, 2014

2402CA303

MACHINE LEARNING LABORATORY

L	T	P	C
0	0	4	2

Prerequisite Data Analytics

COURSE OBJECTIVES

1. To understand about data cleaning and data preprocessing
2. To familiarize with the Supervised Learning algorithms and implement them in practical situations
3. To familiarize with unsupervised Learning algorithms and carry on the implementation part
4. To involve the students to practice ML algorithms and techniques
5. Learn to use algorithms for real time data sets

COURSE OUTCOMES

On the successful completion of the course, students will be able to

- CO1 Apply data preprocessing techniques for predictive modeling.
CO2 Train and evaluate machine learning models.
CO3 Implement feature selection and dimensionality reduction.
CO4 Construct Bayesian networks for probabilistic analysis.
CO5 Apply machine learning algorithms to real-world problems.

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	3	2	3	1	-	-	2	2	3
CO2	3	3	3	3	1	-	-	2	3	3
CO3	2	3	3	3	1	-	-	2	3	3
CO4	2	3	3	3	-	-	2	2	3	3
CO5	3	3	3	3	2	1	1	3	3	3
Avg	2.6	3	2.8	3	1.25	1	1.5	2.2	2.8	3

List of experiments:

1. Demonstrate how do you structure data in Machine Learning
2. Implement data preprocessing techniques on real time dataset
3. Implement Feature subset selection techniques
4. Demonstrate how will you measure the performance of a machine learning model
5. Write a program to implement the naïve Bayesian classifier for a sample training data set. Compute the accuracy of the classifier, considering few test data sets
6. Write a program to construct a Bayesian network considering medical data. Use this model to demonstrate the diagnosis of heart patients using the standard Heart Disease Data Set
7. Apply EM algorithm to cluster a set of data stored in a .CSV file
8. Write a program to implement k-Nearest Neighbor algorithm to classify the data set
9. Apply the technique of pruning for a noisy data monk2 data, and derive the decision tree from this data. Analyze the results by comparing the structure of pruned and unpruned tree
10. Build an Artificial Neural Network by implementing the Backpropagation algorithm and test the same using appropriate data sets
11. Implement Support Vector Classification for linear kernels
12. Implement Logistic Regression to classify problems such as spam detection. Diabetes predictions and so on

TOTAL :60 PERIODS

Software Requirements

Python or any ML tools like R

2402CA304

INTERNET OF THINGS LABORATORY

L T P C
0 0 4 2

Prerequisite Computer Networks, Programming in C / Python

COURSE OBJECTIVES

1. To design applications to interact with sensors
2. To design and develop IoT application Arduino/Raspberry pi for real world scenario
3. To enable communication between IoT and cloud platforms
4. To develop applications using Django Framework

COURSE OUTCOMES

On the successful completion of the course, students will be able to

- CO1 Apply and understand various IoT communication protocols
CO2 Experiment with sensors and actuators for IoT applications.
CO3 Develop IoT applications using Arduino/Raspberry Pi boards.
CO4 Interface sensors and post data to cloud platforms.
CO5 Build and deploy IoT applications using suitable frameworks and platforms.

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	2	1	3	1	–	–	–	3	2
CO2	3	2	3	2	2	–	–	–	3	2
CO3	3	2	3	3	2	1	–	1	3	3
CO4	3	2	3	3	2	2	–	1	3	3
CO5	3	2	3	3	3	2	1	2	3	3
Avg	3	2	2.6	2.8	2	1.6	1	1.3	3	2.6

List of experiments:

1. To study various IoT protocols – 6LowPAN, IPv4/IPv6, Wifi, Bluetooth, MQTT
2. IoT Application Development Using sensors and actuators (temperature sensor, light sensor, infrared sensor)
3. To study Raspberry Pi development board and to implement LED blinking applications
4. To develop an application to send and receive data with Arduino using HTTP request
5. To develop an application that measures the room temperature and posts the temperature value on the cloud platform
6. To develop an application that measures the moisture of soil and post the sensed data over Google Firebase cloud platform
7. To develop an application for measuring the distance using ultrasonic sensor and post distance value on Google Cloud IoT platform
8. Develop a simple application based on sensors
9. Develop IoT applications using Django Framework and Firebase/ Bluemix platform
10. Develop a commercial IoT application

TOTAL :60 PERIODS

2402CA305 PROFESSIONAL DEVELOPMENT COURSES – III

Prerequisite Basic Mathematics & English Language Skills

L T P C
0 0 2 1

COURSE OBJECTIVES

1. Develop logical reasoning, decision-making, and data interpretation skills.
2. Enhance employability through ethics, project handling, and personal branding.
3. Strengthen critical reasoning in judgments, arguments, and cause–effect analysis
4. Improve professional communication in speaking and writing.

COURSE OUTCOMES

On the successful completion of the course, students will be able to

CO1 Apply logical reasoning to solve puzzles and data interpretation problems.

CO2 Develop employability and communication skills.

CO3 Analyze situations critically to make sound decisions

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	3	2	2	2	–	–	–	3	2
CO2	2	2	3	2	3	3	2	3	2	3
CO3	3	3	3	3	3	2	2	2	3	3
Avg	2.6	2.6	2.6	2.3	2.6	2.5	2	2.5	2.6	2.6

UNIT I Decision Making

10 Hours

Logical Deduction, Decision making, Arrangements / Puzzles, Data Interpretations, Data Sufficiency

UNIT II Sharpen Employability Skills

10 Hours

Corporate ethics, Project handling, Emotional Intelligence, Habits & Personality builds character, Resume presentation, Video Resume, Brand yourself, Mock interview

UNIT III Critical Reasoning

10 Hours

Making Judgments, Statement and Conclusion, Argument, Assumption, Logical Problems and Games, Essential Part, Matching Definition, Artificial Languages, Cause and Effect.

Practice:

Conversation building, Speaking and Writing

REFERENCE

TOTAL : 30 PERIODS

1. Arun Sharma, _How to Prepare for Logical Reasoning for CAT‘, 4th edition, McGraw Hills publication, 2017.
2. R S Agarwal, _A modern approach to Logical reasoning‘, revised edition, S.Chand publication, 2017.
3. B.S. Sijwalii and Indu Sijwali, —A New Approach to REASONING Verbal & Non-Verball, 2nd edition, Arihnat publication, 2014
4. Objective General English by SP Bakshi.
5. A Modern approach to verbal and non verbal reasoning by R.S. Agarwal.
6. Complete reference campus recruitment book.

Note:

- Credit: 01
- No. of hours – 30
- Mark allocation:
Assignment/Activity – 3 Nos. = 50 Marks
Assessment – 50 Marks

PROFESSIONAL ELECTIVE II						
Course Code	Course Name	L	T	P	C	Category
2403CA007	DevOp sand Microservices	3	0	0	3	PE
2403CA008	Deep Learning	3	0	0	3	PE
2403CA009	Social Network Analytics	3	0	0	3	PE
2403CA010	Crypto currency and Blockchain Technologies	3	0	0	3	PE
2403CA011	Data Science	3	0	0	3	PE
2403CA012	Computer Vision	3	0	0	3	PE

2403CA007

DEVOPS AND MICROSERVICES

L T P C
3 0 0 3

Prerequisite Basic programming and software development knowledge.

COURSE OBJECTIVES

1. To introduce Micro services and Containers
2. To understand the key concepts and principles of DevOps
3. To be familiar with most common DevOps tools
4. To explain the business benefits of DevOps and continuous delivery
5. To recall specific DevOps methodologies and frameworks

COURSE OUTCOMES

On the successful completion of the course, students will be able to

- CO1 Apply microservice design principles to build scalable applications.
CO2 Integrate microservices in a DevOps environment.
CO3 Utilize DevOps tools and practices for automation and testing.
CO4 Develop and deploy applications using CI/CD pipelines.
CO5 Monitor and manage cloud-based DevOps applications.

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	2	2	3	1	-	-	1	3	2
CO2	2	3	3	3	2	1	-	2	3	3
CO3	2	2	3	3	2	2	1	2	3	3
CO4	2	3	3	3	3	3	-	2	3	3
CO5	2	2	3	3	3	2	1	3	3	3
Avg	2.2	2.4	2.8	3	2.2	2	1	2	3	2.8

UNIT I INTRODUCTION TO MICROSERVICES

9 Hours

Definition of Microservices – Characteristics - Microservices and Containers – Interacting with Other Services – Monitoring and Securing the Services – Containerized Services – Deploying on Cloud

UNIT II MICRO SERVICES ARCHITECTURE

9 Hours

Monolithic architecture- Microservices architectural style- Benefits - Drawbacks of Microservices architectural style - decomposing monolithic applications into Microservices

UNIT III DEVOPS TOOLS

9 Hours

History of DevOps- DevOps and Software Development Life Cycle – Waterfall Model _Agile Model – DevOps LifeCycle – DevOps Tools: distributed version of control tool Git- Automation testing tools- Selenium – report generation –TestNG – User Acceptance Testing – Jenkins

UNIT IV MICROSERVICESINDEVOPSENVIRONMENT

9 Hours

Evolution of Microservices and DevOps – Benefits of combining DevOps and Microservices- working of DevOps and Microservices in Cloud environment - DevOps Pipeline representation for a NodeJS based Microservices

UNIT V VELOCITYANDCONTINUOUSDELIVERY

9 Hours

Velocity - Delivery Pipeline- test stack - Small/Unit Test – medium /integration testing –system testing- Job of Development and DevOps- Job of Test and DevOps– Job of Op and Devops- Infrastructure and the job of Ops

REFERENCE

TOTAL : 45 PERIODS

1. NamitTanasseri,RahulRai,MicroserviceswithAzure,1stEdition,PacktPublishing, UK,2017
2. EberhardWolff,Microservices:FlexibleSoftwareArchitecture,1stEdition,Pearson Education,2017
3. JamesAScott,APracticalGuidetoMicroservicesandContainers,MapRData Technologies e – book.
<https://mapr.com/ebook/microservices-and->
4. Joyner Joseph, Devops for Beginners, First Edition, MihailsKonoplovs publisher, 2015
5. Gene Kim, Kevin Behr, George Spafford, The Phoenix Project, A Novel about IT, DevOps, 5th Edition, IT Revolution Press, 2018
6. Michael Hüttermann, DevOps for Developers, 1st Edition, APress, e-book, 2012

2403CA008

DEEP LEARNING

L	T	P	C
3	0	0	3

Prerequisite Basic programming and software development knowledge.

COURSE OBJECTIVES

1. Develop and Train Deep Neural Networks
2. Develop a CNN, R-CNN, Fast R-CNN, Faster R-CNN, Mask-RCNN for detection and recognition
3. Build and train RNNs, work with NLP and Word Embeddings
4. The internal structure of LSTM and GRU and the differences between them
5. The Auto Encoders for Image Processing

COURSE OUTCOMES

- On the successful completion of the course, students will be able to
- CO1 Feature Extraction from Image and Video Data
- CO2 Implement Image Segmentation and Instance Segmentation in Images
- CO3 Implement image recognition and image classification using a pretrained network (Transfer Learning)
- CO4 Traffic Information analysis using Twitter Data
- CO5 Auto encoder for Classification & Feature Extraction

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	2	2	3	3	1	–	–	1	3	3
CO2	2	3	3	3	1	–	–	1	3	3
CO3	2	2	3	3	1	–	–	1	3	3
CO4	1	2	2	3	2	–	1	2	2	3
CO5	2	2	3	3	1	–	–	2	3	3
Avg	1.8	2.2	2.8	3	1.2	-	1	1.4	2.8	3

UNIT I DEEP LEARNING CONCEPTS

9 Hours

Fundamentals about Deep Learning. Perception Learning Algorithms. Probabilistic modelling. Early Neural Networks. How Deep Learning different from Machine Learning. Scalars. Vectors. Matrixes, Higher Dimensional Tensors. Manipulating Tensors. Vector Data. Time Series Data. Image Data. Video Data

UNIT II NEURAL NETWORKS

9 Hours

About Neural Network. Building Blocks of Neural Network. Optimizers. Activation Functions. Loss Functions. Data Pre-processing for neural networks, Feature Engineering. Overfitting and Underfitting. Hyperparameters.

UNIT III CONVOLUTIONAL NEURAL NETWORK

9 Hours

About CNN. Linear Time Invariant. Image Processing Filtering. Building a convolutional neural network. Input Layers, Convolution Layers. Pooling Layers. Dense Layers. Backpropagation Through the Convolutional Layer. Filters and Feature Maps. Backpropagation Through the Pooling Layers. Dropout Layers and Regularization. Batch Normalization. Various Activation Functions. Various Optimizers. LeNet, AlexNet, VGG16, ResNet. Transfer Learning with Image Data. Transfer Learning using Inception Oxford VGG Model, Google Inception Model, Microsoft ResNet Model. RCNN, Fast R-CNN, Faster R-CNN, Mask-RCNN, YOLO

UNIT IV NATURAL LANGUAGE PROCESSING USING RNN

9 Hours

About NLP & its Toolkits. Language Modeling . Vector Space Model (VSM). Continuous Bag of Words (CBOW). Skip-Gram Model for Word Embedding. Part of Speech (PoS) Global Co occurrence Statistics-based Word Vectors. Transfer Learning. Word2Vec. Global Vectors for Word Representation GloVe. Backpropagation Through Time. Bidirectional RNNs (BRNN). Long Short Term Memory (LSTM). Bi-directional LSTM. Sequence-to-Sequence Models (Seq2Seq). Gated recurrent unit GRU

UNIT V DEEP REINFORCEMENT & UNSUPERVISED LEARNING

9 Hours

About Deep Reinforcement Learning. Q-Learning. Deep Q-Network (DQN). Policy Gradient Methods. Actor-Critic Algorithm. About Autoencoding. Convolutional Auto Encoding. Variational 35 Auto Encoding. Generative Adversarial Networks. Autoencoders for Feature Extraction. Auto Encoders for Classification. Denoising

Auto encoders Sparse Autoencoders

REFERENCE

TOTAL : 45 PERIODS

1. Deep Learning A Practitioner's Approach Josh Patterson and Adam Gibson O'Reilly Media, Inc. 2017
2. Learn Keras for Deep Neural Networks ,Jojo Moolayil, Apress, 2018
3. Deep Learning Projects Using TensorFlow2, Vinita Silaparasetty, Apress, 2020
4. Deep Learning with Python, FRANÇOIS HOLLET, MANNING/SHELTER ISLAND, 2017
5. Pro Deep Learning with TensorFlow, Santanu Pattanayak, Apress, 2017

2403CA009

SOCIAL NETWORK ANALYTICS

L T P C
3 0 0 3

Prerequisite Data Mining

COURSE OBJECTIVES

1. To gain knowledge about social networks, its structure and their data sources
2. To study about the knowledge representation technologies for social network analysis.
3. To analyze the data left behind in social networks
4. To gain knowledge about the community-maintained social media resources
5. To learn about the visualization of social networks

COURSE OUTCOMES

On the successful completion of the course, students will be able to

CO1 create entities and relationships of data as network and do analysis

CO2 Model and represent knowledge for social semantic Web

CO3 Use extraction and mining tools for analyzing Social networks

CO4 Collect data from various social media resources and analyse

CO5 Develop personalized visualization for Social networks

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	2	3	3	3	1	–	–	1	3	3
CO2	2	2	3	3	1	–	–	1	3	2
CO3	2	3	3	3	1	–	1	2	3	3
CO4	1	2	3	3	2	–	1	2	3	3
CO5	1	2	2	3	1	–	–	2	2	2
Avg	1.6	2.4	2.8	3	1.2	-	1	1.6	2.8	2.6

UNIT I INTRODUCTION TO SEMANTICWEB

9 Hours

The development of Semantic Web –Emergence of the Social Web –The Development of Social Network Analysis – Basic Graph Theoretical Concepts of Social Network Analysis – Electronic Sources for Network Analysis–Electronic Discussion Networks, Blogs and Online Communities

UNIT II KNOWLEDGE REPRESENTATION ON THE SEMANTIC WEB

9 Hours

Ontology-based knowledge Representation – Ontology languages for the Semantic Web: RDF and OWL

UNIT III SOCIAL NETWORKMINING

9 Hours

Detecting Communities in Social Network – Evaluating Communities –Methods for Community Detection – Applications of Community Mining Algorithms – Tools for detecting communities – Application: Mining Facebook

UNIT IV COMMUNITY MAINTAINED SOCIAL MEDIA RESOURCES

9 Hours

Community Maintained Resources–Supporting technologies for community maintained resources– User motivations- Location based social interaction –location technology–mobile location sharing–Automated recommender system

UNIT V VISUALIZATION OF SOCIAL NETWORKS

9 Hours

Visualization of Social Networks - Node-Edge Diagrams – Random Layout – Force-Directed Layout – Tree Layout – Matrix Representations –Matrix and Node-Link Diagrams– Visualizing Online Social Networks

REFERENCE

TOTAL : 45 PERIODS

1. Matthew A. Russell, "Mining the Social Web: Data Mining Facebook, Twitter, LinkedIn, Google+, Github and more", O'REILLY, Third Edition, 2018.
2. Charu Aggarwal, "Social Network Data Analytics", Springer, First Edition, 2014
3. Jennifer Golbeck, "Analyzing the social web", Waltham, MA: Morgan Kaufmann (Elsevier), First Edition, 2013.
4. Borko Furht, "Handbook of Social Network Technologies and Applications", Springer, First Edition, 2010
5. Peter Mika, "Social Networks and the Semantic Web", Springer, First Edition, 2007

2403CA010 CRYPTOCURRENCY AND BLOCKCHAIN TECHNOLOGIES L T P C
Prerequisite Computer Network 3 0 0 3

COURSE OBJECTIVES

1. To understand the basics of Cryptocurrency
2. To understand the working of digital tokens and wallets
3. To understand the working of contracts
4. To understand the working of block chain platforms

COURSE OUTCOMES

On the successful completion of the course, students will be able to

- CO1 Identify Blockchain as Data structure and Distribution Data
CO2 Implement the transactions of Crypto currency
CO3 Identify the different ways to achieve Blockchain Technology
CO4 Design and build smart contracts
CO5 Use smart contract for real world application in a Blockchain Platform

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	2	2	2	3	1	–	1	1	3	2
CO2	2	3	3	3	1	–	1	1	3	3
CO3	2	2	3	3	1	–	1	1	3	3
CO4	1	2	3	3	2	–	1	2	3	3
CO5	1	2	3	3	2	–	1	2	3	3
Avg	1.2	1.8	2.4	2.4	1.2	-	1	1.2	2.4	2.4

UNIT I OVERVIEW OF BLOCKCHAIN: 9 Hours
Why Blockchain - The Structure of Blockchain - Data Structure of Blockchain - Data Distribution in Blockchain - Block Validation. Block Validators: Consensus - Proof of Work – Proof of Stake - Proof of Activity - Proof of Elapsed Time - Proof of Burn

UNIT II CRYPTOCURRENCY 9 Hours
Bitcoin: Bitcoin Working - Bitcoin Transactions - Bitcoin Mining - Value of Bitcoin - Community, Politics and Regulations – Advantages – Disadvantages. Ethereum: Overview – Decentralized Application. Components of Ethereum: Smart contracts – Ether - Ethereum Clients – Ethereum Virtual Machine– Ether scripter

UNIT III DEVELOPMENT FRAMEWORKS 9 Hours
Digital Tokens: Overview - Initial Coin Offering – OmiseGO – EOS – Tether. Meta Mask: Wallet Seed – Meta Mask Transactions. Mist: Overview - Mist wallet. Truffle: Features of Truffle – Development Truffle boxes - Community truffle box

UNIT IV HYPERLEDGER 9 Hours
Hyperledger Fabric: Introduction - Fabric v/s Ethereum – Hyperledger Iroha - Features of Iroha. Hyperledger Sawtooth: Components of sawtooth - Proof of Elapsed time

UNIT V BLOCKCHAIN PLATFORMS 9 Hours
Multichain - HydraChain. Future Blockchain: IOTA – Corda - Chain Core. Blockchain Framework: CoCo Framework – Tierion – BigchainDB

REFERENCE **TOTAL : 45 PERIODS**

1. Josh Thompson, 'Blockchain: The Blockchain for Beginnings, Guide to Blockchain Technology and Blockchain Programming', Create Space Independent Publishing Platform, 1st Edition, 2017.
2. Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller, and Steven Goldfeder. Bitcoin and cryptocurrency technologies: a comprehensive introduction. 1st Edition, Princeton University Press, 2016
3. Joseph Bonneau et al, SoK: Research perspectives and challenges for Bitcoin and cryptocurrency, IEEE Symposium on security and Privacy, 1st Edition, 2015
4. Antony Lewis, The Basics of Bitcoins and Blockchains: An Introduction to Cryptocurrencies and the Technology that Powers Them, Mango Publishing group, 2018
5. Tiana Laurence, Introduction to Blockchain Technology, 1st Edition, Van Haren Publishing, 2019.

2403CA011

DATA SCIENCE

L T P C
3 0 0 3

Prerequisite Data Mining

COURSE OBJECTIVES

- 1 To know the fundamental concepts of data science and analytics.
- 2 To learn fundamental data analysis using R.
- 3 To understand various data modeling techniques
- 4 To learn the basic and advanced features of opensource bigdata tools and frameworks.
- 5 To study various analytics on stream data.

COURSE OUTCOMES

- On the successful completion of the course, students will be able to
- CO1 Understand data science concepts and perform exploratory data analysis.
- CO2 Apply R programming for univariate, bivariate, and multivariate data analysis.
- CO3 Build statistical and regression models with effective visualizations.
- CO4 Utilize Hadoop ecosystem tools for big data storage and processing.
- CO5 Apply stream analytics techniques for real-time data processing.

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	2	2	1	1	–	–	1	2	2
CO2	3	2	3	2	1	–	–	1	3	2
CO3	2	2	3	3	1	1	–	1	3	3
CO4	2	3	3	3	2	1	1	1	3	3
CO5	1	2	3	3	2	1	1	1	3	3
Avg	2.2	2.2	2.8	2.4	1.4	1	1	1	2.8	2.6

UNIT I INTRODUCTION TO DATA SCIENCE AND BIG DATA

9 Hours

Introduction to Data Science– Data Science Process–Exploratory Data analysis– Bigdata: Definition, Risks of Big Data, Structure of Big Data – Web Data: The Original Big Data – Evolution Of Analytic Scalability– Analytic Processes and Tools– Analysis versus Reporting – Core Analytics versus Advanced Analytics– Modern Data Analytic Tools – Statistical Concepts: Sampling Distributions – Re-Sampling – Statistical Inference – Introduction to Data Visualization

UNIT II DATA ANALYSIS USING R

9 Hours

Univariate Analysis: Frequency, Mean, Median, Mode, Variance, Standard Deviation, Skewness and Kurtosis – Bivariate Analysis: Correlation – Regression Modeling: Linear and Logistic Regression – Multivariate Analysis – Graphical representation of Univariate, Bivariate and Multivariate Analysis in R: Bar Plot, Histogram, Box Plot, Line Plot, Scatter Plot, Lattice Plot, Regression Line, Two-Way cross Tabulation.

UNIT III DATA MODELING

9 Hours

Univariate Analysis: Frequency, Mean, Median, Mode, Variance, Standard Deviation, Skewness and Kurtosis – Bivariate Analysis: Correlation – Regression Modeling: Linear and Logistic Regression – Multivariate Analysis – Graphical representation of Univariate, Bivariate and Multivariate Analysis in R: Bar Plot, Histogram, Box Plot, Line Plot, Scatter Plot, Lattice Plot, Regression Line, Two-Way cross Tabulation.

UNIT IV DATA ANALYTICAL FRAMEWORKS

9 Hours

Introduction to Hadoop: Hadoop Overview – RDBMS versus Hadoop – HDFS (Hadoop Distributed File system): Components and Block Replication – Introduction to MapReduce – Running Algorithms Using MapReduce – Introduction to HBase: HBase Architecture, HLog and HFile, Data Replication – Introduction to Hive, Spark and Apache Sqoop

UNIT V STREAM ANALYTICS

9 Hours

Introduction To Streams Concepts – Stream Data Model and Architecture – Stream Computing – Sampling

Data in a Stream– Filtering Streams– Counting Distinct Elements in a Stream – Estimating Moments – Counting Oneness in a Window – Decaying Window.

REFERENCE

TOTAL : 45 PERIODS

1. Joel Grus, "Data Science from Scratch: First Principles with Python", O'Reilly Media First edition (April 30, 2015) [eBook available]
2. Aurelien Geron, "Hands-On Machine Learning with Scikit-Learn and Tensor Flow: Concepts, Tools, and Techniques to Build Intelligent Systems", 1st Edition, 2017, O'Reilly Media.
3. Stephen Marsland, Machine learning: An Algorithmic Perspective, CRC Press, Second Edition, 2009.
4. G. Strang (2016) Introduction to Linear Algebra, Wellesley-Cambridge Press, Fifth edition, USA.
5. Ian Goodfellow, Yoshua Bengio and Aaron Courville, "Deep Learning", MIT Press, First Edition (November 18, 2016)
6. Montgomery, D. C. and G. C. Runger (2011). Applied Statistics and Probability for Engineers. 5th Edition. John Wiley & Sons, Inc., NY, USA.

2403CA012

COMPUTER VISION

Prerequisite Basic Python and linear algebra knowledge

L	T	P	C
3	0	0	3

COURSE OBJECTIVES

- 1 To review image processing techniques for computer vision.
- 2 To understand shape and region analysis.
- 3 To understand Hough Transform and its applications to detect lines, circles, ellipses
- 4 To understand three-dimensional image analysis techniques
- 5 To understand motion analysis.
- 6 To study some applications of computer vision algorithms

COURSE OUTCOMES

- On the successful completion of the course, students will be able to
- CO1 Implement fundamental image processing techniques required for computer vision
- CO2 Perform shape analysis.
- CO3 Apply Hough Transform for line, circle, and ellipse detections.
- CO4 Apply 3D vision techniques.
- CO5 Develop applications using computer vision techniques

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	3	2	3	1	-	-	1	3	2
CO2	3	3	2	3	1	-	-	1	3	2
CO3	3	3	3	3	1	-	-	1	3	2
CO4	3	3	3	3	1	-	-	1	3	2
CO5	3	3	3	3	2	1	1	2	3	3
Avg	3	3	2.6	3	1.2	1	1	1.2	3	2.2

UNIT I IMAGE PROCESSING FOUNDATIONS

9 Hours

Review of image processing techniques – classical filtering operations – thresholding techniques – edge detection techniques – corner and interest point detection – mathematical morphology – texture.

UNIT II SHAPES AND REGIONS

9 Hours

Binary shape analysis – connectedness – object labeling and counting – size filtering – distance functions – skeletons and thinning – deformable shape analysis – boundary tracking procedures – active contours – shape models and shape recognition – centroidal profiles – handling occlusion – boundary length measures – boundary descriptors – chain codes – Fourier descriptors – region descriptors – moments.

UNIT III HOUGH TRANSFORM

9 Hours

Line detection – Hough Transform (HT) for line detection – foot-of-normal method – line localization – line fitting – RANSAC for straight line detection – HT based circular object detection – accurate center location – speed problem – ellipse detection – Case study: Human Iris location – hole detection – generalized Hough Transform (GHT) – spatial matched filtering – GHT for ellipse detection – object location – GHT for feature collation.

UNIT IV 3D VISION AND MOTION

9 Hours

Methods for 3D vision – projection schemes – shape from shading – photometric stereo – shape from texture – shape from focus – active range finding – surface representations – point-based representation – volumetric representations – 3D object recognition – 3D reconstruction – introduction to motion – triangulation – bundle adjustment – translational alignment – parametric motion – spline-based motion – optical flow – layered motion.

UNIT V APPLICATIONS

9 Hours

Application: Photo album – Face detection – Face recognition – Eigen faces – Active appearance and 3D shape

models of faces Application: Surveillance – foreground-background separation – particle filters – Chamfer matching, tracking, and occlusion – combining views from multiple cameras – human gait analysis
Application: In-vehicle vision system: locating roadway – road markings – identifying road signs – locating pedestrians.

REFERENCE

TOTAL : 45 PERIODS

1. D. L. Baggio et al., —Mastering OpenCV with Practical Computer Vision Projects, Packt Publishing, 2012.
2. E. R. Davies, —Computer & Machine Vision, Fourth Edition, Academic Press, 2012
3. Jan Erik Solem, —Programming Computer Vision with Python: Tools and algorithms for analyzing images, O'Reilly Media, 2012
4. Mark Nixon and Alberto S. Aquado, —Feature Extraction & Image Processing for Computer Vision, Third Edition, Academic Press, 2012
5. R. Szeliski, —Computer Vision: Algorithms and Applications, Springer 2011.
6. Simon J. D. Prince, —Computer Vision: Models, Learning, and Inference, Cambridge University Press, 2012

PROFESSIONAL ELECTIVE III						
CourseCode	Course Name	L	T	P	C	Category
2403CA013	Cyber security	3	0	0	3	PE
2403CA014	Parallel Computing	3	0	0	3	PE
2403CA015	Agile Methodologies	3	0	0	3	PE
2403CA016	Cognitive Computing	3	0	0	3	PE
2403CA017	Natural Language Processing	3	0	0	3	PE
2403CA018	Network Programming	3	0	0	3	PE

2403CA013

CYBERSECURITY

L T P C
3 0 0 3

Prerequisite **Mathematical Foundations of Computer Applications, Computer Communication and Networks**

COURSE OBJECTIVES

- 1 To learn the principles of cyber security and to identify threats and risks
- 2 To learn how to secure physical assets and develop system security controls
- 3 To understand how to apply security for Business applications and Network communications
- 4 To learn the technical means to achieve security
- 5 To learn to monitor and audit security measures

COURSE OUTCOMES

- On the successful completion of the course, students will be able to
- CO1 Develop and implement risk and security requirements to eliminate gaps in organizational security practices
- CO2 Apply management, operational, and technical controls for effective cyber security
- CO3 Conduct security audits and monitor cyber security performance.
- CO4 Identify gaps and propose improvements in existing security systems.
- CO5 Detect and report system vulnerabilities using appropriate assessment tools.

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	3	3	3	1	1	2	2	3	3
CO2	3	3	3	3	2	2	2	2	3	3
CO3	3	3	3	3	1	2	2	2	3	3
CO4	3	3	3	3	1	1	2	2	3	3
CO5	3	3	3	3	1	1	2	2	3	3
Avg	3	3	3	3	1.2	1.4	2	2	3	3

- UNIT I PLANNING FOR CYBERSECURITY** 9 Hours
Best Practices-Standards and a plan of Action-Security Governance Principles, components and Approach-Information Risk Management-Asset Identification-Threat Identification-Vulnerability Identification-Risk Assessment Approaches-Likelihood and Impact Assessment-Risk Determination, Evaluation and Treatment-Security Management Function-Security Policy-Acceptable Use Policy- Security Management Best Practices - Security Models: Bell La Padula model, Biba Integrity Model - Chinese Wall model
- UNIT II SECURITY CONTROLS** 9 Hours
People Management-Human Resource Security-Security Awareness and Education-Information Management- Information Classification and handling-Privacy-Documents and Record Management-Physical Asset Management-Office Equipment-Industrial Control Systems-Mobile Device Security-System Development-Incorporating Security into SDLC-Disaster management and Incident response planning
- UNIT III CYBERSECURITY FOR BUSINESS APPLICATIONS AND NETWORKS** 9 Hours
Business Application Management-Corporate Business Application Security-End user Developed Applications-System Access- Authentication Mechanisms-Access Control-System Management- Virtual Servers-Network Storage Systems-Network Management Concepts-Firewall-IP Security-Electronic Communications -Case study on OWASP vulnerabilities using OWASP ZAP tool
- UNIT IV TECHNICAL SECURITY** 9 Hours
Supply Chain Management-Cloud Security-Security Architecture-Malware Protection-Intrusion Detection-Digital Rights Management-Cryptographic Techniques-Threat and Incident Management- Vulnerability Management-Security Event Management-Forensic Investigations-Local Environment Management-Business Continuity.
- UNIT V SECURITY ASSESSMENT** 9 Hours

Security Monitoring and Improvement-Security Audit-Security Performance-Information Risk
Reporting-Information Security Compliance Monitoring-Security Monitoring and Improvement Best Practices
REFERENCE TOTAL : 45 PERIODS

1. D. L. Baggio et al., —Mastering OpenCV with Practical Computer Vision Projects, Packt Publishing, 2012.
2. E. R. Davies, —Computer & Machine Vision, Fourth Edition, Academic Press, 2012
3. Jan Erik Solem, —Programming Computer Vision with Python: Tools and algorithms for analyzing images, O'Reilly Media, 2012
4. Mark Nixon and Alberto S. Aquado, —Feature Extraction & Image Processing for Computer Vision, Third Edition, Academic Press, 2012
5. R. Szeliski, —Computer Vision: Algorithms and Applications, Springer 2011.
6. Simon J. D. Prince, —Computer Vision: Models, Learning, and Inference, Cambridge University Press, 2012

2403CA014

PARALLEL COMPUTING

L	T	P	C
3	0	0	3

Prerequisite Basics of Computer Architecture and Operating Systems.

COURSE OBJECTIVES

- 1 To understand the concepts Parallel Computers, Data and Temporal Parallelism
- 2 To learn Structures of Parallel Computers
- 3 To understand the concepts of Operating Systems for Parallel Computers
- 4 To acquire knowledge on CUDA.

COURSE OUTCOMES

On the successful completion of the course, students will be able to

- CO1 Solve the Problems in Parallel
- CO2 Have knowledge on Different Structures of Parallel Computers.
- CO3 Understand the Performance Evaluation of Parallel Computers.
- CO4 Get acquaintance on CUDA.
- CO5 Develop Parallel Programs In CUDA C.

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	3	3	3	1	-	-	2	3	3
CO2	3	3	2	3	1	-	-	1	3	2
CO3	3	3	3	3	1	1	1	2	3	3
CO4	3	3	3	3	1	-	-	2	3	3
CO5	3	3	3	3	1	1	-	2	3	3
Avg	3	3	2.8	3	1	1	1	1.8	3	2.8

UNIT I INTRODUCTION

9 Hours

High Speed Computing - Increase the Speed of Computers - History of Parallel Computers - Solving problems in parallel: Utilizing Temporal Parallelism - Utilizing Data Parallelism - Comparison of Temporal and Data Parallel Processing - Data Parallel Processing with Specialized Processors.

UNIT II STRUCTURE OF PARALLEL COMPUTERS

9 Hours

Generalized Structure of a Parallel Computer - Classification of Parallel Computers - Vector Computers - Typical Vector Super Computer - Array Processors - Shared Memory Parallel Computers - Distributed Shared Memory Parallel Computers - Message Passing Parallel Computers

UNIT III OPERATING SYSTEMS FOR PARALLEL COMPUTERS

9 Hours

Resource Management - Process Management - Process Synchronization - Inter-process Communication - Memory Management - Input/output(Disk Arrays) - Basics of Performance Evaluation - Performance Measurement Tools.

UNIT IV COMPUTER UNIFIED DEVICE ARCHITECTURE

9 Hours

The age of parallel processing - The rise of GPU computing – CUDA - Applications of CUDA - Development Environment-CUDA Enabled Graphics Processors - NVIDIA Device driver - CUDA Development Tool kit - Standard C compiler.

UNIT V CUDA C

9 Hours

Introduction to CUDA C: First program - Querying Devices - Using Device Properties - Parallel Programming in CUDA C: CUDA Parallel Programming - Summing Vectors program

REFERENCE

TOTAL : 45 PERIODS

TEXT BOOKS:

- 1.Parallel Computers Architecture and Programming, V. Rajaraman, C. Siva Ram Murthy, PHI
- 2.CUDA By Example, Jason Sanders, Edward Kandrot, Addison_Wesley.

REFERENCES

- 1.Introduction to Parallel Computing, Ananth Grama, Anshul Gupta, George Karypis, Vipin Kumar, Pearson Education
- 2.Parallel Computing Theory and Practice, Michel j.Quinn

2403CA015

AGILE METHODOLOGIES

L	T	P	C

Prerequisite Software Engineering basics.

3 0 0 3

COURSE OBJECTIVES

- 1 To provide students with a theoretical as well as practical understanding of agile software development practices and how small teams can apply them to create high-quality software
- 2 To provide a good understanding of software design and a set of software technologies and APIs
- 3 To do a detailed examination and demonstration of Agile development and testing techniques
- 4 To understand the benefits and pitfalls of working in an Agile team
- 5 To understand Agile development and testing

COURSE OUTCOMES

- On the successful completion of the course, students will be able to
- CO1 Understand Agile principles and values.
- CO2 Apply Agile process frameworks like Scrum and XP.
- CO3 Use Agile knowledge management practices.
- CO4 Handle Agile-based requirements engineering.
- CO5 Apply Agile QA, metrics, and Scrum techniques.

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	2	1	1	1	1	–	1	2	1	–
CO2	3	2	2	1	1	–	–	2	2	1
CO3	2	2	1	2	1	–	–	1	2	1
CO4	3	2	2	1	1	–	–	1	2	1
CO5	3	2	2	1	2	–	–	2	2	1
Avg	2.6	1.8	1.6	1.2	1.2	-	1	1.6	1.8	1

- UNIT I AGILE FUNDAMENTAL** 9 Hours
Theories for Agile Management – Agile Software Development – Traditional Model vs. Agile Model - Classification of Agile Methods – Agile Manifesto and Principles – Agile Project Management – Agile Team Interactions – Ethics in Agile Teams - Agility in Design, Testing – Agile Documentations – Agile Drivers, Capabilities and Values
- UNIT II AGILE PROCESSES** 9 Hours
Lean Production -SCRUM, Crystal, Feature Driven Development-Adaptive Software Development - Extreme Programming:MethodOverview–Lifecycle–WorkProducts,RolesandPractices
- UNIT III AGILITY AND KNOWLEDGE MANAGEMENT** 9 Hours
Agile Information Systems – Agile Decision Making –Development, Acquisition, Refinement, Distribution, Deployment, Leveraging – KM in Software Engineering – Managing Software Knowledge – Challenges of Migrating to Agile Methodologies – Agile Knowledge Sharing – Role of Story-Cards – Story-Card Maturity Model (SMM)
- UNIT IV AGILITY AND REQUIREMENTS ENGINEERING** 9 Hours
Impact of Agile Processes in RE–Current Agile Practices – Variance – Overview of RE Using Agile – Managing Unstable Requirements – Requirements Elicitation – Agile Requirements Abstraction Model – Requirements Management in Agile Environment,– Agile Requirements Modeling and Generation
- UNIT V AGILITY AND QUALITY ASSURANCE** 9 Hours
Agile Product Development – Agile Metrics – Feature Driven Development (FDD) – Financial and Production Metrics in FDD – Agile Approach to Quality Assurance - Agile Approach in GlobalSoftware Development-Agile Scrum-Scrum Master–Scaling Projects using Scrum

REFERENCE

TOTAL : 45 PERIODS

1. David J. Anderson and Eli Schragenheim,, “Agile Management for Software Engineering: Applying the Theory of Constraints for Business Results”, Illustrated Edition, Prentice HallPTR,2004
2. OritHazzaand Yaepl Dubinsky,“ Agile Software Engineering,: Undergraduate Topic sin Computer Science, SpringerVerlag, FirstEdition,2009

3. Craig Larman, "Agile and Iterative Development: A Manager's Guide", Pearson Education, Second Impression, 2007
4. Kevin C. Desouza, "Agile Information Systems: Conceptualization, Construction, and Management", Elsevier, Butterworth-Heinemann, First Edition, 2007
5. Ken Schwaber, "Agile Project Management with Scrum", Illustrated, Revised Edition Microsoft Press, 2004
6. Konnor Cluster, "Agile Project Management: Learn How To Manage a Project With Agile Methods, Scrum, Kanban and Extreme Programming", Independently Published, First Edition, 2019

2403CA016

COGNITIVE COMPUTING

Prerequisite Artificial Intelligence Basics, Neural Networks

L	T	P	C
3	0	0	3

COURSE OBJECTIVES

- 1 To know the theoretical background of cognition
- 2 To understand the link between cognition and computational intelligence.
- 3 To explore probabilistic programming language
- 4 To study the computational inference models of cognition.
- 5 To study the computational inference models of cognition.

COURSE OUTCOMES

- On the successful completion of the course, students will be able to
- CO1 Understand the underlying theory behind cognition.
- CO2 Design the cognition elements computationally.
- CO3 Implement mathematical functions through WebPPL.
- CO4 Develop a cognitive inference model.
- CO5 Develop a cognitive learning model.

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	2	1	2	1	–	1	1	1	–
CO2	3	3	2	2	2	–	–	1	2	1
CO3	3	3	2	3	3	–	–	1	3	2
CO4	3	3	3	3	2	–	–	1	3	2
CO5	3	3	3	3	2	–	–	1	3	2
Avg	3	2.8	2.2	2.6	2	-	1	1	2.4	1.7

UNIT I PHILOSOPHY, PSYCHOLOGY AND NEUROSCIENCE 9 Hours

Philosophy: Mental-physical Relation – From Materialism to Mental Science – Detour before the naturalistic turn – The Philosophy of Science – The Mind in Cognitive Science – Logic and the Sciences of the Mind – Psychology: Place of Psychology within Cognitive Science – Science of Information Processing – Neurosciences: Cognitive Neuroscience – Perception – Decision – Learning and Memory – Language Understanding and Processing

UNIT II COMPUTATIONAL INTELLIGENCE 9 Hours

Machines and Cognition – Artificial Intelligence – Architectures of Cognition – Knowledge Based Systems – Logical Representation and Reasoning – Logical Decision Making – Decision making under Uncertainty – Learning – Language – Vision – Robotics.

UNIT III PROBABILISTIC PROGRAMMING LANGUAGE 9 Hours

Web PPL Language – Syntax – Using JavaScript Libraries – Manipulating probability types and distributions – Finding Inference – Exploring random computation – Coroutines: Functions that receive continuations – Enumeration – Other basic computation.

UNIT IV IMPLEMENTING THE INFERENCE MODELS OF COGNITION 9 Hours

Generative Models – Conditioning – Causal and statistical dependence – Conditional dependence – Data Analysis – Algorithms for Inference

UNIT V IMPLEMENTING THE LEARNING MODELS OF COGNITION 9 Hours

Learning as Conditional Inference – Learning with a Language of Thought – Hierarchical Models – Occam's Razor – Learning (Deep) Continuous Functions – Mixture Models.

REFERENCE

TOTAL : 45 PERIODS

1. Robert A. Wilson, Frank C. Keil, "The MIT Encyclopedia of the Cognitive Sciences", The MIT Press, 1999.
2. Noah D. Goodman, Andreas Stuhlmüller, "The Design and Implementation of Probabilistic Programming Languages", Electronic version of book, <https://dippl.org/>
3. Benjamin B. Bederson and Ben Shneiderman, "The Craft of Information Visualization" Morgan Kaufmann Publishers, 2003.
4. Noah D. Goodman, Joshua B. Tenenbaum, The ProbMods Contributors, "Probabilistic
5. Models of Cognition", Second Edition, 2016, <https://probmods.org/>.
6. <https://cognitivecomputingconsortium.com/Applications>, Second Edition, A.K. Peters/ CRC Press, 2015.

2403CA017

NATURAL LANGUAGE PROCESSING

L	T	P	C

Prerequisite	Programming Fundamentals, Data Structures and Algorithms	3	0	0	3
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COURSE OBJECTIVES

- 1 To understand basics of linguistics, probability and statistics
- 2 To study statistical approaches to NLP and understand sequence labeling
- 3 To outline different parsing techniques associated with NLP
- 4 To explore semantics of words and semantic role labeling of sentences
- 5 To understand discourse analysis, question answering and chatbots

COURSE OUTCOMES

On the successful completion of the course, students will be able to

CO1	Understand basics of linguistics, probability and statistics associated with NLP
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CO2 Implement a Part-of-Speech Tagger

C03	Design and implement a sequence labeling problem for a given domain
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CO4	Implement semantic processing tasks and simple document indexing and searching system using the concepts of NLP
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CO5 Implement a simple chatbot using dialogue system concepts

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	2	1	1	–	–	–	2	1	2
CO2	3	3	2	2	1	–	–	2	3	2
CO3	3	3	3	2	1	–	–	2	3	3
CO4	3	3	3	3	1	1	–	2	3	3
CO5	2	2	3	3	2	1	1	3	3	2
Avg	2.8	2.6	2.4	2.2	1.25	1	1	2.2	2.6	2.4

UNIT I	INTRODUCTION
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9 Hours

Natural Language Processing – Components - Basics of Linguistics and Probability and Statistics – Words-Tokenization-Morphology-Finite State Automata

UNIT II STATISTICAL NLP AND SEQUENCE LABELING

9 Hours

N-grams and Language models –Smoothing -Text classification- Naïve Bayes classifier – Evaluation - Vector Semantics
– TF-IDF - Word2Vec- Evaluating Vector Models -Sequence Labeling – Part of Speech – Part of Speech Tagging -
Named Entities –Named Entity Tagging

UNIT III CONTEXTUAL EMBEDDING

9 Hours

Constituency – Context Free Grammar – Lexicalized Grammars- CKY Parsing – Earley's algorithm Evaluating Parsers - Partial Parsing – Dependency Relations- Dependency Parsing - Transition Based - Graph Based

UNIT IV COMPUTATIONAL SEMANTICS

9 Hours

Word Senses and WordNet – Word Sense Disambiguation – Semantic Role Labeling – Proposition Bank- FrameNet- Selectional Restrictions - Information Extraction - Template Filling

UNIT V DISCOURSE ANALYSIS AND SPEECH PROCESSING

9 Hours

Discourse Coherence – Discourse Structure Parsing – Centering and Entity Based Coherence – Question Answering – Factoid Question Answering – Classical QA Models – Chatbots and Dialogue systems – Frame-based Dialogue Systems – Dialogue-State Architecture

REFERENCE

TOTAL : 45 PERIODS

1. Daniel Jurafsky and James H. Martin, "Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition" (Prentice Hall Series in Artificial Intelligence), 2020
2. Jacob Eisenstein. "Natural Language Processing", MIT Press, 2019
3. Samuel Burns "Natural Language Processing: A Quick Introduction to NLP with Python and NLTK", 2019
4. Christopher Manning, "Foundations of Statistical Natural Language Processing", MIT Press, 2009.
5. Nitin Indurkha, Fred J. Damerau, "Handbook of Natural Language Processing", Second edition, Chapman & Hall/CRC: Machine Learning & Pattern Recognition, Hardcover, 2010
6. Deepti Chopra, Nisheeth Joshi, "Mastering Natural Language Processing with Python", Packt Publishing Limited, 2016

7. Mohamed Zakaria Kurdi “Natural Language Processing and Computational Linguistics: Speech, Morphology and Syntax (Cognitive Science)”, ISTE Ltd., 2016
8. Atefeh Farzindar, Diana Inkpen, “Natural Language Processing for Social Media (Synthesis Lectures on Human Language Technologies)”, Morgan and Claypool Life Sciences, 2015

2403CA018

NETWORK PROGRAMMING

L T P C

Prerequisite Programming Fundamentals, Operating Systems Basics

3 0 0 3

COURSE OBJECTIVES

- 1 To understand the basics of Network Programming
- 2 To be familiar with building network applications
- 3 To design and implement client server Applications using TCP and UDP Sockets
- 4 To expose with various socket options
- 5 To get aware of Network security for Network Programming

COURSE OUTCOMES

On the successful completion of the course, students will be able to

- CO1 Design and implement the client/server programs using variety of protocols
- CO2 Understand the key protocols which support Internet
- CO3 Demonstrate advanced knowledge of programming interfaces for network communication
- CO4 Use the basic tools for design and testing of network programs in Unix environment
- CO5 Identify some of the factors driving the need for network security

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	3	3	2	1	–	–	2	3	1
CO2	3	2	1	1	–	–	–	2	1	1
CO3	3	3	3	3	1	–	–	2	3	2
CO4	3	2	2	3	1	1	–	2	3	1
CO5	2	2	1	2	–	–	3	2	1	–
Avg	2.8	2.4	2	2.2	1	1	3	2	2.2	1.25

UNIT I INTRODUCTION

9 Hours

TCP/IP Layer Model – Multicast, broadcast and Anycast - Socket address Structures – Byte ordering functions – address conversion functions – Elementary TCP Sockets – socket, connect, bind, listen, accept, read, write, close functions – Iterative Server – Concurrent Server

UNIT II UNIT TITLE

9 Hours

TCP Echo Server – TCP Echo Client – Posix Signal handling – Server with multiple clients – boundary conditions: Server process Crashes, Server host Crashes, Server Crashes and reboots, Server Shutdown

UNIT III SOCKET OPTIONS AND MULTIPLEXING

9 Hours

Socket options – get socket and set socket functions – generic socket options – IP socket options – ICMP socket options – TCP socket options I/O multiplexing – I/O Models – select function – shutdown function – TCP echo Server (with multiplexing) – poll function – TCP echo Client (with Multiplexing)

UNIT IV ELEMENTARY UDP SOCKETS

9 Hours

UDP echo Server – UDP echo Client – Multiplexing TCP and UDP sockets – Domain name system – get host by name function – Ipv6 support in DNS – get host by adr function – get serv by name and get serv by port functions

UNIT V NETWORK SECURITY

9 Hours

SSL - SSL Architecture, SSL Protocols, SSL Message, Secure Electronic Transaction (SET). TLS – TLS Protocols, DTLS Protocols, PKI – Fundamentals, Standards and Applications

REFERENCE

TOTAL : 45 PERIODS

1. W. Richard Stevens, Bill Fenner, Andrew M. Rudoff, "Unix Network Programming, Volume The Sockets Networking API", Third Edition, ISBN:0-13-141155-1, Addison Wesley Pearson Education, 2004

2. Behrouz A. Forouzan, Debdeep Mukhopadhyay “Cryptography and Network Security”
Second Edition, ISBN-13: 978-0-07-070208-0 Tata McGraw Hill Education Private Limited 2010
3. William Stallings, “Cryptographic and network security Principles and Practices”, Fourth Edition, Publisher Prentice Hall, November 2005
4. Andre Perez, ” Network Security ”, First Edition, Publisher John Wiley & Sons, 2014
5. Gary R. Wright , W. Richard Stevens, ”TCP/IP Illustrated: The Implementation” , ISBN 0-201-63354-X , Vol. 2, 1st Edition , Addison Wesley Professional, January 2008

PROFESSIONAL ELECTIVE IV						
CourseCode	Course Name	L	T	P	C	Category
2403CA019	R Programming	3	0	2	4	PE
2403CA020	Generative AI for Software development	3	0	2	4	PE
2403CA021	Advanced Java Programming	3	0	2	4	PE
2403CA022	C#and.NET	3	0	2	4	PE
2403CA023	Mixed Reality	3	0	2	4	PE
2403CA024	Web Design	3	0	2	4	PE

2403CA019

R Programming

L T P C

Prerequisite Problem Solving and Programming

3 0 2 4

COURSE OBJECTIVES

- 1 To understand the basics in R programming in terms of constructs, control statements, string functions
- 2 To understand the use of R for Big Data analytics
- 3 To Learn to apply R programming for Text processing
- 4 To explore the R programming from a statistical perspective

COURSE OUTCOMES

On the successful completion of the course, students will be able to

- CO1 Design programs using R data structure constructs.
- CO2 Develop programs using R Matrices and Arrays.
- CO3 Construct programs using R Data Frames
- CO4 Design graphics applications using oop concepts.
- CO5 Apply interfacing rules in combining R scripts with other languages.

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	2	1	2	–	–	–	2	2	2
CO2	3	2	2	2	–	–	–	2	2	3
CO3	3	3	2	2	–	–	–	2	3	3
CO4	3	2	3	3	1	–	–	2	3	2
CO5	2	2	3	3	–	–	–	3	3	2
Avg	2.8	2.2	2.2	2.4	1	–	–	2.2	2.6	2.4

UNIT I INTRODUCTION

15 Hours

Introducing to R – R Data Structures – Help functions in R – Vectors – Scalars – Declarations – recycling – Common Vector operations – Using all and any – Vectorized operations – NA and NULL values – Filtering – Vectorised if-then else – Vector Equality – Vector Element names

Lab component:

1. Installation of R Studio and package exploration
2. Program samples on R data structures and Vectors.

UNIT II MATRICES, ARRAYS AND LISTS

15 Hours

Creating matrices – Matrix operations – Applying Functions to Matrix Rows and Columns – Adding and deleting rows and columns – Vector/Matrix Distinction – Avoiding Dimension Reduction – Higher Dimensional arrays – lists – Creating lists – General list operations – Accessing list components and values – applying functions to lists – recursive lists.

Lab component:

3. Program on Arrays and list demo.
4. Program demonstration of Matrices.

UNIT III DATA FRAMES

15 Hours

Creating Data Frames – Matrix-like operations in frames – Merging Data Frames – Applying functions to Data frames – Factors and Tables – factors and levels – Common functions used with factors – Working with tables – Other factors and table related functions – Control statements – Arithmetic and Boolean operators and values – Default values for arguments – Returning Boolean values – functions are objects – Environment and Scope issues – Writing Upstairs – Recursion – Replacement functions – Tools for composing function code – Math and Simulations in R

Lab component:

1. Implementing R Control Structures.
2. Implementing Data Frames in R.
3. Demonstrating Functions in R

UNIT IV OBJECT ORIENTED PROGRAMMING 15 Hours

S3 Classes – S4 Classes – Managing your objects – Input/Output – accessing keyboard and monitor – reading and writing files – accessing the internet – String Manipulation – Graphics – Creating Graphs – Customizing Graphs – Saving graphs to files – Creating three-dimensional plots

Lab component:

1. String Manipulation demonstration using R.

Creating various graphs in R.

UNIT V INTERFACING 15 Hours

Interfacing R to other languages – Parallel R – Basic Statistics – Linear Model – Generalized Linear models – Non-linear models – Time Series and Auto-correlation – Clustering

Lab component:

R interface examples

FURTHER READING: Visualizations in R

REFERENCE TOTAL : 75 PERIODS

1. The Art of R Programming: A Tour of Statistical Software Design, Norman Matloff, No Starch Press, 2011
2. R for Everyone: Advanced Analytics and Graphics, Jared P. Lander, Addison-Wesley Data & Analytics Series, 2013.
3. Beginning R – The Statistical Programming Language, Mark Gardener, Wiley, 2013
4. Introductory R: A Beginner's Guide to Data Visualisation, Statistical Analysis and Programming in R, Robert Knell, Amazon Digital South Asia Services Inc, 2013
5. <https://www.tutorialspoint.com/r/index.htm>
6. <https://www.r-project.org/about.html>

2403CA020

GENERATIVE AI FOR SOFTWARE DEVELOPMENT

L T P C

Prerequisite Artificial Intelligence

3 0 2 4

COURSE OBJECTIVES

- 1 To explore the concepts of AI Fundamentals
- 2 To introduce the Roles of AI Tools
- 3 To make understand the usage of OpenAI APIS
- 4 To explore the Application Area of AI
5. To provide the knowledge of software development using AI

COURSE OUTCOMES

- On the successful completion of the course, students will be able to
- CO1 Explain the basic concepts of Generating AI
- CO2 Develop Generative AI with Custom Data set using Botsonic, Chatbase
- CO3 Create a website using Gen AI Tools
- CO4: Build Chat GPT powered applications
- CO5: Develop one chatbot building project from various available projects in domains like Travel, E- Commerce. Education, Legal, etc.

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	2	2	1	2	–	–	1	2	1	1
CO2	3	3	3	3	1	–	1	2	3	3
CO3	2	2	3	3	1	–	–	2	3	2
CO4	3	3	3	3	1	–	1	2	3	2
CO5	2	3	3	3	2	1	2	3	3	2
Avg	2.4	2.6	2.6	2.8	1.2	1	1.2	2.2	2.6	2

UNIT I INTRODUCTION TO THE PROGRAM AND FUNDAMENTALS

15 Hours

Introduction to Generative AI-Chat GPT: What is a Prompt-Elements of a Prompt- Designing Prompt- Example prompts for various use cases, Generative Texts-Introduction to AI Chat bots-Working of AI Chat bots-Popular AI Chat bots- Chat GPT and its working-usage of Chat GPT- Use cases of Chat GPT for various users

Lab Components

1. Applications of generative models using text generation
2. Explore integrating Chat GPT with a voice interface using text-to-speech

UNIT II GENERATIVE MODEL FOR TEXT

15 Hours

Generative Texts : Language Models Basics – Building blocks of Language models - Transformer Architecture – Encoder and Decoder – Attention mechanisms - Generation of Text- Generation of Text – Autoencoding – Regression Models – Exploring Chat GPT – Prompt Engineering- Programming LLM

Lab Components

Generate a responsive HTML web page based on a shorted scription (e.g., "A contact form with name, email, and message").
Generate SQL queries from English instructions

UNIT III OPEN AI APIS

15 Hours

Understanding Open AI APIs-Open AI playground – creating API keys - Authentication-making requests- Building Chat GPT powered applications- Open AI APIs- build chat bots- integration with Open AI API keys, Build a website using Gen AI Tools, Generative AI with Custom Data set : Botsonic , Chatbase: Train with custom dataset-Introduction about Gemini AI

Lab Components

1. Installing procedure for chatGPT
2. Create Chatbot using GPT Model

UNIT IV APPLICATION AREA OF AI

15 Hours

From simple perceptron to multi-layer perceptron and its training, Deep learning and its architectures- Introducing deep learning and architectures such as CNN, RNN, GAN and transformers, Transformer and Large Language Model, Training LLM for natural language processing

Lab Components

Solving XOR problem using DNN

Face recognition using CNN

UNIT V SOFTWARE DEVELOPMENT USING AI

15 Hours

Generative Adversarial Networks (GAN) for image generation, Complete one chatbot building project from various available projects in domains like Travel, E-Commerce. Education, Legal, etc

Lab Components

. Image augmentation using GANs

Machine Translation using Encoder-Decoder model

REFERENCE

TOTAL : 75 PERIODS

1. Generative Deep Learning” by David Foster

2. Demystifying Prompt Engineering: A Practical Guide to AI Prompts by Harish Bhat.

3. Rebooting AI : Building Artificial Intelligence we can trust ” by Gary Marcus and Ernest Davis

4. Artificial Intelligence “ A guide for Thinking Humans ” by Melanie Mitchell

5. Natural Language Processing in Action” by Derve

2403CA021

ADVANCED JAVA PROGRAMMING

L T P C
3 0 2 4

Prerequisite Object oriented Programming

COURSE OBJECTIVES

- 1 To understand the fundamentals of web programming and client-side scripting.
- 2 To learn server-side development using Servlets and WebSockets.
- 3 To learn the Spring Framework and build applications using Spring.
- 4 To learn and implement the concepts of Java Persistence API (JPA).
5. To learn advanced client-side scripting and frameworks.

COURSE OUTCOMES

On the successful completion of the course, students will be able to

- CO1 To write client-side scripting.
CO2 To implement the server side of the web application.
CO3 To implement a web application using Spring.
CO4: To implement a Java application using Java Persistence API.
CO5: To implement a full-stack Single Page Application using React, Spring, and JPA.

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	2	2	1	2	–	–	–	2	2	1
CO2	3	3	2	2	1	–	–	2	3	1
CO3	3	3	3	3	1	–	–	2	3	2
CO4	3	3	3	3	–	–	–	2	3	2
CO5	3	3	3	3	2	1	1	3	3	2
Avg	2.8	2.8	2.4	2.6	1.3	1	1	2.2	2.8	1.6

UNIT I INTRODUCTION TO WEB & JAVASCRIPT

15 Hours

Introduction to Web: Server - Client - Communication Protocol (HTTP), Javascript Prototypes - Classes - Modules – Fetch API – JS Canvas - Storage: Local Storage, Cookies, IndexedDB, JSON

Lab Components

1. Create an event registration application using javascript. It should implement different widgets for registration form and registered records view using tabs. It should perform the form validation. (Unit I)
2. Create a javascript application in an Object Oriented way using Classes and Modules. It should also use browser storage for persistence. (Unit I)

UNIT II SERVERSIDE PROGRAMMING

15 Hours

Web Server: Web Containers - Web Components, Servlet: Lifecycle - Request - Servlet Context -Response - Filter - Session - Dispatching Requests, WebSocket, Logging - Log4j2, Build tool - Gradle. Introduction to Spring: IoC Container and Dependency Injection (DI)

Lab Components

- 1 Build a web application using Gradle. The server side of the application should implement RESTful APIs using Servlet and do necessary logging. The client side of the application should be a single page application which consumes the RESTful APIs through AJAX. (Unit II)
2. Build a chat application using WebSocket. (Unit II)

UNIT III SPRING

15 Hours

Spring Configuration and Spring Boot, Spring MVC, Spring Bean Lifecycle – Dispatcher Servlet and Configuration - Interceptors – Annotations, Controllers - Views - Input Validation - File Upload - Container, Dependency and IOC.

Lab Components

1. Create a Spring MVC application. The application should handle form validation, file upload, and session tracking.

3. 2. Implement a RESTful Spring Boot application using Spring REST, Spring Security, and Spring Cache.

UNIT IV AOP, JAVA PERSISTENCE API AND HIBERNATE

15 Hours

Aspect Oriented Programming (AOP) - Entity: Basic, Embeddable and Collection Types – Identifiers - Entity Relationship - Inheritance, Persistence Context and Entity Manager, JPQL, Criteria API, Spring Data JPA - Specification and Projection.

Lab Components

1. Design a system using JPA and Hibernate. The system should have multiple entities and relationships between the entities. The database schema should be generated through Hibernate. Provide RESTful endpoints for CRUD operations for the defined entities. Also, support pagination and searching using JPA's JPQL and Criteria API. (Unit IV)
2. Create a Spring RESTful Application with Spring Data JPA. Support pagination and searching using Specifications. (Unit IV)

UNIT V ADVANCED SPRING PROGRAMMING

15 Hours

Spring Boot JDBC - Spring Boot Actuator - Spring Cloud - Spring Boot Testing - Spring Security Architecture, Spring Cache - Building RESTful Web Services

1. Create a React application with different components and interactions between the components.
2. Develop a full-stack application using React and Spring. Make use of Spring REST, Spring
Develop a full-stack application using React and Spring. Make use of Spring REST, Spring Security, Spring
3. Data JPA, Hibernate, Spring Boot, Gradle and ReactJS state and component mechanism.

REFERENCE

TOTAL : 75 PERIODS

1. David Flanagan, JavaScript: The Definitive Guide, O'Reilly Media, Inc., 7th Edition, 2020.
2. Matt Frisbie, Professional JavaScript for Web Developers, Wiley Publishing, Inc., 4th Edition, ISBN: 978-1-119-36656-0, 2019.
3. Alex Banks and Eve Porcello, Learning React, O'Reilly Media, Inc., 2nd Edition, 2020.
4. David R. Heffelfinger, Java EE 8 Application Development, Packt Publishing, First Edition, 2017.
Benjamin Muschko, Gradle in Action, Manning Publications, First Edition, 2014.
5. Iuliana Cosmina, Rob Harrop, Chris Schaefer, Clarence Ho, Pro Spring 5: An In-Depth Guide to the Spring Framework and Its Tools, Apress, Fifth Edition, 2017.

2403CA022

C# AND.NET

L T P C

Prerequisite PHP

3 0 2 4

COURSE OBJECTIVES

- 1 To learn the technologies of the .NET framework.
- 2 To cover all segments of programming in C# starting from the language basis, followed by the object oriented programming concepts.
- 3 To update and enhance skills in writing Windows applications, ADO.NET and ASP .NET.
- 4 To introduce advanced topics namely data connectivity, WPF, WCF and WPF with C# and .NET 4.5.
5. To implement mobile applications using .Net Compact Framework

COURSE OUTCOMES

On the successful completion of the course, students will be able to

CO1 Understand the difference between .NET and Java framework.

CO2 Work with the basic and advanced features of C# language

CO3 Create applications using various data providers.

CO4: Create a web application using ASP.NET.

CO5: Create mobile applications using .NET compact framework

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	2	2	1	1	–	–	–	2	1	–
CO2	3	2	2	2	–	–	–	2	2	–
CO3	3	3	3	2	1	–	–	2	3	2
CO4	3	3	3	3	1	–	1	2	3	2
CO5	3	2	3	3	1	1	1	3	3	2
Avg	2.8	2.4	2.4	2.2	1	1	1	2.2	2.4	2

UNIT I .NET FRAME WORK INTRODUCTION

15 Hours

.Net Architecture–Core C#–Variables–Data Types–Flow control–Objects and Types–Classes and Structs– Inheritance–Generics–ArraysandTuples–OperatorsandCasts–Indexers–Assemblies–Shared Assemblies– CLR Hosting–App domains

UNIT II C# ADVANCED FEATURES

15 Hours

Delegates – Lambdas – Lambda Expressions – Events – Event Publisher – Event Listener –Strings and Regular Expressions – Generics – Collections – Memory Management and Pointers –Errors and Exceptions–Reflection

UNIT III BASE CLASS LIBRARIES AND DATAMANIPULATION

15 Hours

DiagnosticsTasks–ThreadsandSynchronization–ManipulatingXML–SAXandDOM–Manipulating files and the Registry – Transactions – Data access with ADO.NET: Introduction, LINQ to Entities and the ADO.NET Entity Framework, Querying a Database with LINQ – Creating the ADO.NET Entity Data Model Class Library, Creating a Windows Forms Project – Data Bindings between Controls

UNIT IV WINDOW AND WEB BASED APPLICATIONS

15 Hours

Window Based Applications – Core ASP.NET – ASP.NET Web Forms – Server Controls, Data Binding – ASP.NET State Management, Tracing, Caching, Error Handling, Security, Deployment, User and Custom Controls–Windows Communication Foundation(WCF

UNIT V NET COMPACT FRAMEWORK

15 Hours

Reflection–.NetRemoting–.NetSecurity–Localization–Peer-to-PeerNetworking–BuildingP2PApplications–.NetCompactFramework–CompactEditionDataStores–TestingandDebugging –Optimizing per formance – Packaging and Deployment

TOTAL : 75 PERIODS

List of Experiments

1. Write a program in C# to check whether a number is palindrome or not
2. Design a simple calculator using switch statement in C#
3. Write a program in C# to find the roots of quadratic equation.
4. Using try, catch and finally blocks write a program in C# to demonstrate error handling
5. Write a program in C# to build a class which implements an interface which already exists.
6. Implement linked lists in C# using the existing collections namespace
7. Write a C# program to create a dataset for student details, use grid view to display information.
8. Write a C# program to add new rows and new columns in the above program (student details) and create methods to access the dataset
9. Write an ASP.Net program to display a welcome message in the form when the button is clicked.
Write an ASP.Net program containing a listbox, button, an image and label controls. When the user clicks on an item in the listbox, its image should be displayed in the image control. When the user clicks the button, the cost of the selected item should be displayed in the control

Reference

1. Christian Nagel, Bill Evjen, Jay Glynn, Karli Watson, and Morgan Skinner, Professional C# and .NET 4.5, Wiley, First Edition, 2012.
2. Andrew Troelsen, Pro C# 5.0 and the .NET 4.5 Framework, Apress Publication, First Edition, 2012.
3. Ian Griffiths, Matthew Adams, and Jesse Liberty, Programming C# 4.0, O'Reilly, Sixth Edition, 2010.
4. Andy Wigley, Daniel Moth, and Peter Foot, Mobile Development Handbook, Microsoft Press, 2nd Edition, 2011.
5. Herbert Schildt, C# – The Complete Reference, Tata McGraw-Hill, First Edition, 2010.

2403CA023

MIXED REALITY

L T P C
3 0 2 4

Prerequisite Computer Graphics

COURSE OBJECTIVES

- 1 To learn the technologies of the .NET framework.
- 2 To gain the knowledge of historical and modern overviews and perspectives on virtual reality.
- 3 To learn the fundamentals of sensation, perception, and perceptual training.
- 4 To have the scientific, technical, and engineering aspects of augmented and virtual reality systems.
- 5 To learn the Evaluation of virtual reality from the lens of design.

COURSE OUTCOMES

- On the successful completion of the course, students will be able to
- CO1 Demonstrate knowledge and understanding of VR, AR, and MR concepts, technologies, and applications.
- CO2 Apply design principles and considerations specific to Mixed Reality platforms.
- CO3 Understand interaction design principles in Mixed Reality.
- CO4: Apply software design and implementation skills for Mixed Reality systems.
- CO5: Demonstrate knowledge of the intersection of Mixed Reality and Human-Robot Interaction (HRI).
- CO6: Analyze and evaluate the integration of Mixed Reality technologies and principles into realworld applications

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	2	1	–	1	–	–	–	2	2	2
CO2	2	2	2	3	–	–	–	2	3	2
CO3	2	2	1	2	–	–	–	2	2	2
CO4	3	3	3	3	1	–	–	2	3	3
CO5	2	2	1	2	–	1	1	2	2	2
CO6	3	3	2	2	1	1	1	2	3	3
Avg	2.3	2.1	1.8	2.1	1	1	1	2	2.5	2.3

UNIT I INTRODUCTION TO MIXED REALITY 15 Hours

Introduction to virtual reality (VR), Augmented reality (AR), and Mixed Reality (MR) – History – MR Use cases & Designing for MR platforms – Mixing Virtual with Real - MR hardware and devices –The Input – The output – Optical see-through displays – Eye Tracking- Computer vision for MR

Lab Components

1. Use the primitive objects and apply various projection types by handling camera.
2. Implementation of VRML to draw a house like structure combines different shapes like Cone, Cylinder, Box

UNIT II INTERACTION DESIGN IN MR 15 Hours

Integrating Framework For MR –Embedded MR environment - Tangible Interaction – Auditory Induced presence – Exertion in MR systems – Mixed Interaction in MR

Lab Components

- 1, Develop VR enabled applications using motion trackers and sensors incorporating full haptic interactivity
2. Develop AR enabled applications with interactivity like E learning environment, Virtual walkthroughs and visualization of historic places

UNIT III SOFTWARE DESIGN AND IMPLEMENTATION 15 Hours

Outdoor MR Systems – Multimodal Excitatory Interfaces – Tracking in Mixed Reality – Authoring

Immersive MR – Collaborative AR – Software Engineering Method for MR

UNIT IV MIXED REALITY AND HUMAN-ROBOT INTERACTION

15 Hours

Mixed Reality for Robots – User-centered HRI – Mental Transformation in HRI – Computational Cognitive Modeling – Evaluating the usability of the virtual environment – Security Robot

Lab Components

1. Robot programming and simulation for pick and place
2. Robot programming and simulation for colour identification

UNIT V APPLICATIONS OF MIXED REALITY

15 Hours

MR- MR in healthcare and medical simulations – Teleoperation interface for MR Robot - MR in psychological experimentation – MR in Sports – MR Prototypes to support early creative design – MR Companion Robots

Lab Components

1. Develop simple MR enabled gaming applications

TOTAL: 75 PERIODS

REFERENCES

1. O'Connell, Kharis, —Designing for Mixed Realityl, O'Reilly Media, Inc, 2016.
2. Dubois E, Gray P, Nigay L, —The engineering of mixed reality systemsl Springer Science & Business Media, 2009
3. Wang, Xiangyu, —Mixed reality and human-robot interactionl. Vol. 47. Springer Science & Business Media, 2011.
4. Benyon, D, —Designing Interactive Systems: A Comprehensive Guide to HCI, UX and Interaction Designl, Third Edition, Pearson Education Limited, 2013.

2403CA024

WEBDESIGN

L T P C
3 0 2 4

Prerequisite Web Technologies

COURSE OBJECTIVES

- 1 To understand the concepts and architecture of the World Wide Web
- 2 To understand and practice markup languages
- 3 To understand and practice embedded dynamic scripting on client-side Internet
- 4 Programming To understand and practice web development techniques on client-side

COURSE OUTCOMES

On the successful completion of the course, students will be able to

CO1 Create a basic website using HTML and Cascading Style Sheets.

CO2 Create websites with complex layouts

CO3 Add interactivity to websites using simple scripts

CO4: Design rich client presentation using AJAX

CO5: Add business logic to websites using PHP and databases

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	2	1	1	2	–	–	–	2	2	–
CO2	2	2	2	3	–	–	–	2	2	–
CO3	3	2	2	3	–	–	–	2	3	–
CO4	3	3	3	3	1	–	–	2	3	1
CO5	3	3	3	3	1	1	1	2	3	2
Avg	2.6	2.2	2.2	2.8	1	1	1	2	2.6	1.5

UNIT I INTRODUCTION TO WWW

9+6 Hours

Understanding the working of Internet-Web Application Architecture-Brief history of Internet-Web Standards – W3C-Technologies involved in Web development – Protocols-Basic Principles involved in developing a website-Five Golden Rules of Web Designing

UNIT II UI DESIGN

9+6 Hours

SVG- Iframes - HTML5 Video and Audio tags - CSS Specificity - Box model - Margins, padding and border – Inline and block elements- Structuring pages using Semantic Tags -Positioning with CSS: Positions, Floats, z-index – CSS with CSS Preprocessors: SASS

UNIT III ADVANCED UI WITH CSS3

9+6 Hours

Layouts with CSS Grids Flexbox– Responsive web design with media queries - Advanced CSS Effects – Gradients, opacity, box-shadow - CSS3 Animations: Transforms and Transitions - CSS Frameworks: Bootstrap

UNIT IV JAVASCRIPT

9+6 Hours

JavaScript Events - Modifying CSS of elements using JavaScript- Javascript Classes- Introduction to JQuery – JQuery Selectors -Using JQuery to add interactivity - JQuery Events-Modifying CSS with JQuery -Adding and removing elements with JQuery-AJAX with JQuery-Animations with JQuery (hide, show, animate, fade methods, Slide Method)

UNIT V SERVER-SIDE PROGRAMMING WITH PHP

9+6 Hours

PHP basic syntax-PHP Variables and basic data structures-Using PHP to manage form submissions-File Handling -Cookies and Sessions with PHP-Working with WAMP and PHPMYADMIN-Establishing connectivity with MySQL using PHP

1. Design a landing page for a website using Adobe XD (Unit 1)
2. Design an Admin Dashboard for an E-commerce website using Adobe XD (Unit 1)
3. Design and develop an event registration form. (Unit 2)
4. Design and develop a sticky navbar using floats and SASS. (Unit 2)
5. Design and develop a developer portfolio page. Develop the layout using flexbox and ensure the page is responsive. (Unit 3)
6. Design and develop pricing card list which are responsive using plain CSS and Flexbox (Unit 3)
7. Develop a register form and validate it using JavaScript. Design the forms using CSS3 and display Error Messages in the HTML page. (Unit 4)
8. Develop a website that uses the 'jsonplaceholder' Api to get posts data and display them in the form of a card Use Flexbox to style the cards (Unit 4)
9. Develop a php server that Creates, Reads, Updates and Deletes To do and save them in MySQL database. (Unit 5)
10. Develop a php server that registers and authenticates user session and stores user data in MySQL database. (Unit 5)

REFERENCES

TOTAL : 75 PERIODS

1. David Flanagan, JavaScript: The Definitive Guide, 7th Edition, O'Reilly Publications, 2020.
2. Danny Goodman, Dynamic HTML: The Definitive Reference – A Comprehensive Resource for XHTML, CSS, DOM, and JavaScript, O'Reilly Publications, 3rd Edition, 2007.
3. Robin Nixon, Learning PHP, MySQL, JavaScript & CSS: A Step-by-Step Guide to Creating Dynamic Websites, O'Reilly Publications, 2nd Edition, 2018.
4. David Sawyer McFarland, CSS: The Missing Manual, O'Reilly Publications, 4th Edition, 2015.
5. Keith J. Grant, CSS in Depth, Manning Publications, 1st Edition, 2018.
6. Elizabeth Castro, HTML5 & CSS3 Visual Quickstart Guide, Peachpit Press, 7th Edition, 2012.
7. Harvey Deitel and Abbey Deitel, Internet and World Wide Web: How to Program, 5th Edition, Pearson Education, 2012.
8. MDN Web Docs, <https://developer.mozilla.org/en-US>

OPEN ELECTIVE						
CourseCode	Course Name	L	T	P	C	Category
2403CA025	Data Visualization in Business Analytics	3	0	0	3	OE
2403CA026	New Product Development	3	0	0	3	OE
2403CA027	Sustainable Management	3	0	0	3	OE
2403CA028	Micro and Small Business Management	3	0	0	3	OE
2403CA029	Intellectual Property Rights	3	0	0	3	OE
2403CA030	IoT for Smart Systems	3	0	0	3	OE

2403CA025

DATA VISUALIZATION IN BUSINESS ANALYTICS

L T P C
3 0 0 3

Prerequisite Database Management Systems, Fundamentals of Statistics

COURSE OBJECTIVES

- 1 To understand the categories of data quality principles.
- 2 To describe data through visual representation
- 3 To provide basic knowledge about how large datasets are represented into visual graphics and easily understand the complex relationships within the data
- 4 To design effective visualization techniques for any different problems

COURSE OUTCOMES

On the successful completion of the course, students will be able to

- CO1 Describe the principles of visual perception
- CO2 Apply visualization techniques for analyzing one-dimensional, two-dimensional, and three-dimensional data.
- CO3 Design visualization techniques by applying appropriate design steps and principles
- CO4: Design information dashboards by applying visual design principles and usability practices
- CO5: Apply visualization systems based on data and analysis types for various applications

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	2	2	2	2	1	1	1	2	2	2
CO2	3	3	3	3	1	1	1	2	3	3
CO3	3	3	3	3	1	1	1	2	3	3
CO4	3	3	3	3	1	1	1	2	3	3
CO5	3	3	3	3	1	1	1	2	3	3
AVG	2.8	2.8	2.8	2.8	1	1	1	2	2.8	2.8

UNIT I INTRODUCTION 9 Hours

Visualization – visualization process – role of cognition – Pseudocode conventions – Scatter plot -Data foundation : Types of data - Structure within and between records - Data preprocessing – Human perceptions and information processing

UNIT II VISUALIZATION FOUNDATIONS 9 Hours

Semiology of graphical Symbols – Eight Visual Variables – Historical Perspective- Visualization Techniques for spatial data – One-dimensional data- two dimensional data – Three dimensional data- dynamic data – combining techniques- Visualization of Geospatial data – Visualization of Point, line, area data

UNIT III DESIGNING EFFECTIVE VISUALIZATION 9 Hours

Steps in Designing Visualization –problemsinDesigningEffectiveVisualization–Comparingandevaluating visualization techniques – Visualization Systems

UNIT IV INFORMATION DASHBOARD DESIGN 9 Hours

Characteristics of dashboards –Key goals in visual design process –Dashboard display media – Designing dashboards for usability– Meaningful organization – Maintaining consistency – Aesthetics of dashboards –Testing for usability – Case Studies: Sales dashboard, Marketing analysis dashboard

UNIT V VISUALIZATION SYSTEMS 9 Hours

Systems based on Data type-systems based on Analysis type – Text analysis and visualization – Modern integrated visualization systems – toolkit-Research directions in visualization – issues of cognition, perception and reasoning – issues of evaluation - issues of Hardware

REFERENCES

TOTAL : 45 PERIODS

1. Matthew O. Ward, Georges Grinstein, and Daniel Keim, Interactive Data Visualization: Foundations, Techniques, and Applications, CRC Press, 2nd Edition, 2015.

2. Stephen Few, Now You See It: Simple Visualization Techniques for Quantitative Analysis, 1st Edition, Analytics Press, 2009.
3. Stephen Few, Information Dashboard Design: The Effective Visual Communication of Data, 1st Edition, O'Reilly Media, 2006.
4. Ben Fry, Visualizing Data: Exploring and Explaining Data with the Processing Environment, 1st Edition, O'Reilly Media, 2013.
5. Andy Kirk, Data Visualization: A Handbook for Data-Driven Design, 2nd Edition, Sage Publications, India, 2019.
6. Claus O. Wilke, Fundamentals of Data Visualization, 1st Edition, O'Reilly Media, USA, 2019.

2403CA026

NEW PRODUCT DEVELOPMENT

L T P C
3 0 0 3

Prerequisite Principles of Marketing

COURSE OBJECTIVES: The main learning objective of this course is to prepare the students for

- 1 Applying the principles of generic development process; and understanding the organization structure for new product design and development.
- 2 Identifying opportunity and planning for new product design and development.
- 3 Conducting customer need analysis; and setting product specification for new product design and development
- 4 Generating, selecting, and testing the concepts for new product design and development
- 5 Applying the principles of Industrial design and prototype for new product design and development.

COURSE OUTCOMES

On the successful completion of the course, students will be able to

- CO1 Apply the principles of generic development process; and understand the organization structure for new product design and development
- CO2 Identify opportunity and plan for new product design and development.
- CO3 Conduct customer need analysis; and set product specification for new product design development.
- CO4: Generate, select, and test the concepts for new product design and development
- CO5: Apply the principles of Industrial design and prototype for design and develop new products

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	2	2	1	1	2	1	1	2	1
CO2	2	3	2	1	1	2	1	1	2	1
CO3	2	3	2	1	1	1	1	1	2	2
CO4	2	2	3	2	1	1	1	1	3	1
CO5	2	2	3	3	2	1	1	1	3	1
Avg	3	2	2	1	1	2	1	1	2	1

UNIT I INTRODUCTION TO PRODUCT DESIGN & DEVELOPMENT 9 Hours

Introduction – Characteristics of Successful Product Development – People involved in Product Design and Development – Duration and Cost of Product Development – The Challenges of Product Development – The Product Development Process – Concept Development: The Front-End Process – Adapting the Generic Product Development Process – Product Development Process Flows – Product Development Organizations

UNIT II OPPORTUNITY IDENTIFICATION & PRODUCT PLANNING 9 Hours

Opportunity Identification: Definition – Types of Opportunities – Tournament Structure of Opportunity Identification – Effective Opportunity Tournaments – Opportunity Identification Process – Product Planning: Four types of Product Development Projects – The Process of Product Planning

UNIT III BAYESIAN LEARNING 9 Hours

Identifying Customer Needs: The Importance of Latent Needs – The Process of Identifying Customer Needs. Product Specifications: Definition – Time of Specifications Establishment – Establishing Target Specifications – Setting the Final Specifications

UNIT IV ARAMETRIC MACHINE LEARNING 9 Hours

Concept Generation: Activity of Concept Generation – Structured Approach – Five step method of Concept Generation. Concept Selection: Methodology – Concept Screening and Concepts Scoring. Concept testing: Seven Step activities of concept testing.

UNIT V INDUSTRIAL DESIGN & PROTOTYPING

9 Hours

Industrial Design: Need and Impact – Industrial Design Process. Prototyping – Principles of Prototyping – Prototyping Technologies – Planning for Prototypes.

TOTAL : 45 PERIODS

TEXTBOOK

Ulrich K.T., Eppinger S. D. and Anita Goyal, —Product Design and Development —McGraw-Hill Education; 7 edition, 2020

REFERENCE

1. Belz A., 36-Hour Course: —Product Development|| McGraw-Hill, 2010
2. Rosenthal S.,—Effective Product Design and Development||, Business One Orwin, Homewood, 1992, ISBN1- 55623-603-4.
3. Pugh.S,—Total Design Integrated Methods for Successful Product Engineering||, Addison Wesley Publishing, 1991, ISBN0-202-41639-5
4. Chitale, A. K. and Gupta, R. C., Product Design and Manufacturing, PHI Learning, 2013
5. Jamnia, A., Introduction to Product Design and Development for Engineers, CRC Press, 2018.

2403CA027

SUSTAINABLE MANAGEMENT

L T P C
3 0 0 3

Prerequisite Principles of Marketing

COURSE OBJECTIVES: The main learning objective of this course is to prepare the students for

- 1 Applying the principles of generic development process; and understanding the organization structure for new product design and development.
- 2 Identifying opportunity and planning for new product design and development.
- 3 Conducting customer need analysis; and setting product specification for new product design and development
- 4 Generating, selecting, and testing the concepts for new product design and development
- 5 Applying the principles of Industrial design and prototype for new product design and development.

COURSE OUTCOMES

On the successful completion of the course, students will be able to

- CO1 Apply the principles of generic development process; and understand the organization structure for new product design and development
- CO2 Identify opportunity and plan for new product design and development.
- CO3 Conduct customer need analysis; and set product specification for new product design development.
- CO4: Generate, select, and test the concepts for new product design and development
- CO5: Apply the principles of Industrial design and prototype for design and develop new products

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	2	2	1	1	1	2	–	1	1	–
CO2	1	3	2	1	1	2	–	1	1	–
CO3	1	3	2	1	1	1	–	1	1	1
CO4	1	2	3	2	2	1	–	1	2	–
CO5	1	1	3	3	2	1	1	2	3	–
Avg	1.2	2.2	2.2	1.6	1.4	1.4	1	1.2	1.6	1

UNIT I INTRODUCTION TO PRODUCT DESIGN & DEVELOPMENT 9 Hours

Introduction – Characteristics of Successful Product Development – People involved in Product Design and Development – Duration and Cost of Product Development – The Challenges of Product Development – The Product Development Process – Concept Development: The Front-End Process – Adapting the Generic Product Development Process – Product Development Process Flows – Product Development Organizations

UNIT II OPPORTUNITY IDENTIFICATION & PRODUCT PLANNING 9 Hours

Opportunity Identification: Definition – Types of Opportunities – Tournament Structure of Opportunity Identification – Effective Opportunity Tournaments – Opportunity Identification Process – Product Planning: Four types of Product Development Projects – The Process of Product Planning

UNIT III BAYESIAN LEARNING 9 Hours

Identifying Customer Needs: The Importance of Latent Needs – The Process of Identifying Customer Needs. Product Specifications: Definition – Time of Specifications Establishment – Establishing Target Specifications – Setting the Final Specifications

UNIT IV ARAMETRICMACHINELEARNING

9 Hours

Concept Generation: Activity of Concept Generation – Structured Approach – Five step method of Concept Generation. Concept Selection: Methodology – Concept Screening and Concepts Scoring. Concept testing: Seven Step activities of concept testing.

UNIT V INDUSTRIALDESIGN&PROTOTYPING

9 Hours

Industrial Design: Need and Impact–Industrial Design Process. Prototyping – Principles of Prototyping – Prototyping Technologies – Planning for Prototypes.

TOTAL : 45 PERIODS

TEXTBOOK

Ulrich K.T., Eppinger S. D. and Anita Goyal, —Product Design and Development—McGraw-Hill Education; 7 edition, 2020

REFERENCE

1. Belz A., 36-Hour Course: —Product Developmentll McGraw-Hill, 2010
2. Rosenthal S.,—Effective Product Design and Developmentll, Business One Orwin,Homewood, 1992,ISBN1- 55623-603-4.
3. Pugh.S,—Total Design Integrated Methods for Successful Product Engineeringll,Addison Wesley
4. Publishing,1991,ISBN0-202-41639-5 Chitale, A. K. and Gupta, R. C., Product Design and Manufacturing, PHI Learning, 2013
5. Jamnia, A., Introduction to Product Design and Development for Engineers, CRC Press, 2018.

2403CA028

MICRO AND SMALL BUSINESS MANAGEMENT

L T P C
3 0 0 3

Prerequisite Management Principles,

COURSE OBJECTIVES: The main learning objective of this course is to prepare the students for

- 1 To familiarize students with the theory and practice of small business management.
- 2 To learn the legal issues faced by small business and how they impact operations.

COURSE OUTCOMES

On the successful completion of the course, students will be able to

- CO1 Familiarize the students with the concept of small business
CO2 In depth knowledge on small business opportunities and challenges
CO3 Ability to devise plans for small business by building the right skills and marketing strategies
CO4: Identify the funding source for small start ups
CO5: Business evaluation for buying and selling of small firms

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	2	1	–	–	–	–	–	1	–	–
CO2	2	3	1	–	–	1	–	1	–	–
CO3	1	2	3	1	2	2	–	1	1	–
CO4	1	2	2	–	–	3	–	1	–	–
CO5	1	3	2	1	–	2	1	1	–	–
Avg	1.4	2.2	2	1	2	2	1	1	1	–

UNIT I INTRODUCTION TO SMALL BUSINESS 9 Hours

Creation, Innovation, entrepreneurship and small business - Defining Small Business –Role of Owner – Manager – government policy towards small business sector –elements of entrepreneurship –evolution of entrepreneurship –Types of Entrepreneurship – social, civic, corporate - Business life cycle - barriers and triggers to new venture creation – process to assist start ups – small business and family business.

UNIT II SCREENING THE BUSINESS OPPORTUNITY AND FORMULATING THE BUSINESS PLAN 9 Hours

Concepts of opportunity recognition; Key factors leading to new venture failure; New venture screening process; Applying new venture screening process to the early stage small firm Role-planning in small business – importance of strategy formulation – management skills for small business creation and development.

UNIT III BUILDING THE RIGHT TEAM AND MARKETING STRATEGY 9 Hours

Management and Leadership – employee assessments – Tuckman’s stages of group development- The entrepreneurial process model - Delegation and team building - Comparison of HR management in small and large firms - Importance of coaching and how to apply a coaching model. Marketing within the small business - success strategies for small business marketing - customer delight and business generating systems, - market research, - assessing market performance- sales management and strategy - the marketing mix and marketing strategy.

UNIT IV FINANCING SMALL BUSINESS 9 Hours

Main sources of entrepreneurial capital; Nature of 'bootstrap' financing - Difference between cash and profit - Nature of bank financing and equity financing - Funding-equity gap for small firms. Importance of working capital cycle - Calculation of break-even point - Power of gross profit margin- Pricing for profit - Credit policy issues and relating these to cash flow management and profitability.

UNIT V VALUING SMALL BUSINESS AND CRISIS MANAGEMENT

9 Hours

Causes of small business failure - Danger signals of impending trouble - Characteristics of poorly performing firms - Turnaround strategies - Concept of business valuation - Different valuation measurements - Nature of goodwill and how to measure it - Advantages and disadvantages of buying an established small firm - Process of preparing a business for sale.

TOTAL : 45 PERIODS

REFERENCE

1. Hankinson,A.(2000). "The key factors in the profile of small firm owner-managers that influence business performance. The South Coast Small Firms Survey, 1997-2000." Industrial and Commercial Training 32(3):94-98.
2. Parker,R.(2000). "Small is not necessarily beautiful: An evaluation of policy support for small and medium-sized enterprise in Australia." Australian Journal of Political Science 35(2):239- 253.Journal articles on SME's.

2403CA029

INTELLECTUAL PROPERTY RIGHTS

L T P C
3 0 0 3

Prerequisite Intellectual Property Management

COURSE OBJECTIVES:

- 1 To understand intellectual property rights and its valuation.

COURSE OUTCOMES

On the successful completion of the course, students will be able to

- CO1 Understanding of intellectual property and appreciation of the need to protect it
CO2 Awareness about the process of patenting
CO3 Understanding of the statutes related to IPR
CO4: Ability to apply strategies to protect intellectual property
CO5: Ability to apply mode ls for making strategic decisions related to IPR

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	2	1	–	–	–	–	–	1	–	–
CO2	2	3	1	–	–	1	–	1	–	–
CO3	1	2	3	1	2	2	–	1	1	–
CO4	1	2	2	–	–	3	–	1	–	–
CO5	1	3	2	1	–	2	1	1	–	–
Avg	1.4	2.2	2	1	2	2	1	1	1	1.4

UNIT I INTRODUCTION

9 Hours

Intellectual property rights - Introduction, Basic concepts, Patents, Copyrights, Trademarks, Trade Secrets, Geographic Indicators; Nature of Intellectual Property, Technological Research, Inventions and Innovations, History - the way from WTO to WIPO, TRIPS.

UNIT II PROCESS

9 Hours

New Developments in IPR, Procedure for grant of Patents, TM, GIs, Patenting under Patent Cooperation Treaty, Administration of Patent system in India, Patenting in foreign countries.

UNIT III STATUTES

9 Hours

International Treaties and conventions on IPRs, The TRIPs Agreement, PCT Agreement, The Patent Act of India, Patent Amendment Act (2005), Design Act, Trademark Act, Geographical Indication Act, Bayh- Dole Act and Issues of Academic Entrepreneurship.

UNIT IV STRATEGIES IN INTELLECTUAL PROPERTY

9 Hours

Strategies for investing in R&D, Patent Information and databases, IPR strength in India, Traditional Knowledge, Case studies.

UNIT V MODELS

9 Hours

The technologies Know-how, concept of ownership, Significance of IP in Value Creation, IP Valuation and IP Valuation Models, Application of Real Option Model in Strategic Decision Making, Transfer and Licensing.

TOTAL : 45 PERIODS

REFERENCES

- 1.V.Sople Vinod, Managing Intellectual Property by (Prentice hall of India Pvt.Ltd), 2006.
2. Intellectual Property rights and copy rights ,Ess Publications.
- 3.Primer, R.Anita Rao and Bhanoji Rao, Intellectual Property Rights, Lastain Book company.
4. Edited by Derek Bosworth and Elizabeth Webster, The Management of Intellectual Property, Edward Elgar Publishing Ltd., 2006. WIPO Intellectual Property Handbook

403CA030

IoT FOR SMART SYSTEMS

L T P C
3 0 0 3

Prerequisite Fundamentals of Business and Economics

COURSE OBJECTIVES: The main learning objective of this course is to prepare the students for:

- 1 To familiarize students with the theory and practice of small business management.
To learn the legal issues faced by small business and how they impact operations

COURSE OUTCOMES

Upon completion of this course, the students will be able to

- CO1 Familiarise the students with the concept of small business
CO2 In depth knowledge on small business opportunities and challenges
CO3 Ability to devise plans for small business by building the right skills and marketing strategies
CO4: Identify the funding source for small start ups
CO5: Business evaluation for buying and selling of small firms

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	2	1	–	–	–	–	–	1	–	–
CO2	2	3	1	1	–	–	–	1	–	–
CO3	2	2	3	3	2	–	–	1	3	1
CO4	2	2	2	3	1	–	1	1	2	2
CO5	1	3	2	2	1	–	1	2	2	2
Avg	1.8	2.2	2	2.25	1.3	--	1	1.2	2.3	1.6

UNIT I INTRODUCTION TO INTERNET OF THINGS 9 Hours

Overview, Hardware and software requirements for IOT, Sensor and actuators, Technology drivers, Business drivers, Typical IoT applications, Trends and implications..

UNIT II IOT ARCHITECTURE 9 Hours

IoT reference model and architecture -Node Structure - Sensing, Processing, Communication, Powering, Networking - Topologies, Layer/Stack architecture, IoT standards, Cloud computing for IoT, Bluetooth, Bluetooth Low Energy beacons

UNIT III PROTOCOLS AND WIRELESS TECHNOLOGIES FOR IOT PROTOCOLS 9 Hours

NFC, SCADA and RFID, Zigbee MIPI, M-PHY, UniPro, SPMI, SPI, M-PCIe GSM, CDMA, LTE, GPRS, small cell **Wireless technologies for IoT:** WiFi (IEEE 802.11), Bluetooth/Bluetooth Smart, ZigBee/ZigBee Smart, UWB (IEEE 802.15.4), 6LoWPAN, Proprietary systems-Recent trends

UNIT IV IOT PROCESSORS 9 Hours

Services/Attributes: Big-Data Analytics for IOT, Dependability, Interoperability, Security, Maintainability.
Embedded processors for IOT :Introduction to Python programming -Building IOT with RASPBERRY PI and Arduino

UNIT V CASE STUDIES 9 Hours

Industrial IoT, Home Automation, smart cities, Smart Grid, connected vehicles, electric vehicle charging, Environment, Agriculture, Productivity Applications, IOT Defense

TOTAL : 45 PERIODS

REFERENCES

- 1.Hankinson,A.(2000). “The key factors in the profile of small firm owner-managers that influence business performance. The South Coast Small Firms Survey, 1997-2000.” Industrial and Commercial Training 32(3):94-98.
- 2.Parker,R.(2000). “Small is not necessarily beautiful: An evaluation of policy support for small and medium-sized enterprise in Australia.” Australian Journal of Political Science 35(2):239- 253.
- 3.Journal articles on SME’s.

BRIDGE COURSES

Bridge Courses (1 year duration)						
Course Code	Course Name	L	T	P	C	Category
SEMESTER III						
2401CA005	Object Oriented Programming using C++	3	0	2	4	PC
2401CA006	Introduction to Computer Organization and Operating Systems	3	0	0	3	PC
	TOTAL	6	0	2	7	

2401CA005

PROGRAMMING CONCEPTS USING C AND C++

L	T	P	C
3	0	0	3

PREREQUISITE :

Fundamentals of Data Structures

COURSE OBJECTIVES:

1. To learn the syntax of C.
2. To get exposed to the file processing techniques of C.
3. To get familiarized with the pre-processor directives
- 4.. To understand the OO paradigm
- 5.To be aware of the OO design technique

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	2	2	2	1	—	—	2	3	1
CO2	3	2	2	2	1	—	—	2	3	1
CO3	3	3	2	2	1	—	—	2	3	2
CO4	2	2	3	3	1	—	—	2	3	1
CO5	2	3	3	3	2	—	—	2	3	1
Average	2.6	2.4	2.4	2.4	1.2	—	—	2.0	3.0	1.2

UNIT I BASICS OF C PROGRAMMING

9 Hours

Introduction–Keywords– Identifiers–Basic Data Types in C–Variables–Constants– Input/output Statements in C–Operators in C–Conditional Branching Statements–Iterative Statements–Nested Loops–The Break and Continue Statements –Goto Statement

UNIT II FUNCTIONS, ARRAYS , STRINGS AND POINTERS

9 Hours

Functions-Definitions- Prototypes – Passing Parameters to the Function – Scope of Variables– Storage Classes – Recursive Functions - Arrays – Declaration – Usage – Passing Arrays to Functions – Reading and Writing Strings – String Operations- Pointer Variable Declarations and Initialization – Operators – Relationship between Pointers and Arrays – Arrays of Pointers – Pointers to Functions .

UNIT III ADT, FILES AND PREPROCESSOR DIRECTIVES

9 Hours

Structures-Definition – Initialization–Unions–Bitwise Operators–Enumeration Constants. Introduction to Files – Using Files in C – Read and Write Data with Files - Random Access Files – Types of Pre-processor Directives –#define - #include- Conditional Directives..

UNIT IV OOPS CONCEPTS

9 Hours

Introduction - object-oriented design - object-oriented features- –types and declaration –pointers –arrays - structures– expressions and statements –functions –namespaces and exceptions.

UNIT V INHERITANCE AND POLYMORPHISM

9 Hours

Classes – Constructors – Destructors – Function Overloading - Operator Overloading – Conversions- Derived Class – Virtual Functions –Polymorphism - Abstract Base Class – Multiple Inheritances

TOTAL: 45 HOURS

FURTHER READING:

Java Serialization, RTTI Templates, STL Algorithms and Function Adaptors

COURSE OUTCOMES:

On the successful completion of the course, students will be able to

- CO1:Design and implement C programs for a given problem
- CO2:Develop the C program for the given problem using Structure.
- CO3:Develop the C program for a given problem using dynamic storage allocation.
- CO4:Develop the C++ programs for any given problem using Class.
- CO5:Design and implement C++ programs for any given problem using, Polymorphism and Inheritance

REFERENCES :

1. R.G.Dromey, “How to Solve it by Computer”, Pearson Education, 2007.
2. ReemaThareja, “Programming in C”, OXFORD Higher Education, 2011.
3. Kernigan Brian W., and Dennis M. Ritchie, “The C Programming Language”, Second Edition,Prentice Hall, 1988
4. Ira Pohl, “Object - Oriented Programming using C++”, Second Edition, Pearson Education, 2003.

5. Bjarne Stroustrup, "The C++ Programming Language", Fourth Edition, Addison Wesley, 2013.
6. Dr. Micheal Arock, "Fundamentals of Programming with C", Yes Dee Publication, 2014.
7. Herbert Schildt, "C++: The Complete Reference", Fifth Edition, McGraw Hill, 2015.
8. Stanley B. Lippman, Josée Lajoie, C++ Primer, Fifth Edition, Barbara E. Moo, Pearson Education, 2012.
9. <https://nptel.ac.in/>

2401CA006	INTRODUCTION TO COMPUTER ORGANIZATION AND OPERATING SYSTEMS	L	T	P	C
		3	0	0	3

PREREQUISITE: Basic Hardware Knowledge, Kernel

COURSE OBJECTIVES:

1. To learn the basic structure and operations of a computer.
2. To learn the arithmetic and logic unit and implementation of fixed-point and floating point arithmetic unit.
3. To understand the memory hierarchies, cache memories and virtual memories and to learn the different ways of communication with I/O devices.
4. To understand the basic concepts and functions of Operating Systems
5. To understand Process and various Scheduling Algorithms of OS

UNIT I BASIC STRUCTURE AND ARITHMETIC OPERATIONS 9 Hours

.Functional Units – Basic Operational Concepts – Instructions: Language of the Computer – Operations, Operands – Instruction representation – Logical operations – Decision Making – MIPS Addressing- Arithmetic for Computers

UNIT II PROCESSOR AND CONTROL UNIT 9 Hours

A Basic MIPS Implementation – Building a Datapath – Control Implementation Scheme – Pipelining – Pipelined datapath and control – Handling Data Hazards & Control Hazards – Exceptions.

UNIT III MEMORY & I/O SYSTEMS 9 Hours

Memory Hierarchy - Memory technologies – cache memory – measuring and improving cache performance – virtual memory – Accessing I/O Devices – Interrupts – Direct Memory Access – Bus structure – Interface circuits – USB

UNIT IV OPERATING SYSTEMS OVERVIEW 9 Hours

Operating system overview-objectives and functions, Evolution of Operating System- Operating System Structure- System Calls- Processes – Process Concept, Inter-process Communication

UNIT V PROCESS MANAGEMENT 9 Hours

CPU Scheduling – Scheduling criteria, Scheduling algorithms, Multiple-processors scheduling, Threads Overview – The critical-section problem, Semaphores, Classical problems of synchronization, Critical regions

TOTAL: 45 PERIODS

COURSE OUTCOMES:

On Completion of the course, the student should be able to:

- CO1: Understand the basic structure of computers, operations and instructions.
- CO2: Design arithmetic and logic unit, control unit.
- CO3: Understand the various memory systems and I/O communication.
- CO4: Understand operating system functions, types, system calls
- CO5: Analyze Process and various scheduling algorithms

REFERENCES:

1. David A. Patterson and John L. Hennessy, Computer Organization and Design: The Hardware/Software Interface, Fifth Edition, Morgan Kaufmann/Elsevier

1. Carl Hamacher, Zvonko Vranesic, Safwat Zaky and Naraig Manjikian, Computer Organization and Embedded Systems, Sixth Edition, Tata McGraw Hill, 2012
2. Abraham Silberschatz, Peter Baer Galvin and Greg Gagne- Operating System Concepts, 9th Edition, John Wiley and Sons Inc., 2012.
4. John P. Hayes, Computer Architecture and Organization, Third Edition, Tata McGraw Hill, 2012
5. John L. Hennessey and David A. Patterson, Computer Architecture—A Quantitative Approach, Morgan Kaufmann/Elsevier Publishers, Fifth Edition, 2012.
6. Andrew S. Tanenbaum- Modern Operating Systems, 4th Edition, Pearson Education, 2014.